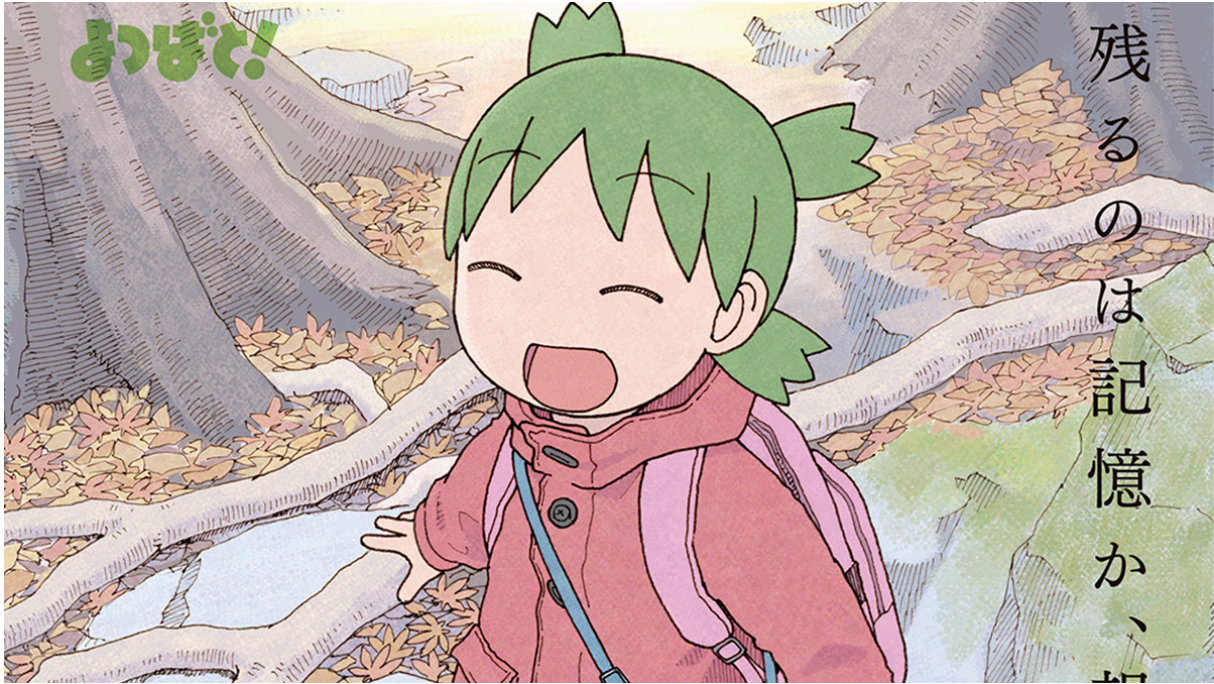


PHYS 161 Notes

Mechanics and Heat

Miagao



Yotsuba from Yotsuba&!

Contents

1	Units, Physical Quantities, and Vectors	4
1.1	SI Units	4
1.2	Unit Conversions	4
1.3	Significant Figures	4
1.4	Vectors	5
	Vector Multiplication	6
2	Motion Along a Straight Line	7
2.1	Displacement, Velocity, and Acceleration	7
2.2	Motion with Constant Acceleration	7
3	Motion in Two or Three Dimensions	9
3.1	Vector Quantities	9
3.2	Projectile Motion	9
3.3	Circular Motion	11
3.4	Relative Motion	12
4	Newton's Laws of Motion	14
4.1	Newton's First Law	14
4.2	Newton's Second Law	14
4.3	Newton's Third Law	14
4.4	Free Body Diagrams	14
5	Applying Newton's Laws	16
5.1	Types of Forces	16
5.2	Forces in Equilibrium	17
5.3	Dynamics of Particles	17
5.4	Dynamics of Circular Motion	19
6	Work and Kinetic Energy	21
6.1	Work	21
6.2	Kinetic Energy	21
6.3	Work and Energy with Varying Force	22
6.4	Power	23
7	Potential Energy and Energy Conservation	24
7.1	Gravitational Potential Energy	24
7.2	Elastic Potential Energy	24
7.3	Conservative and Nonconservative forces	25
	The Law of Conservation of Energy	25
7.4	Force and Potential Energy	29
8	Momentum, Impulse, and Collisions	31
8.1	Momentum and Impulse	31
8.2	Conservation of Momentum	33
8.3	Momentum Conservation and Collisions	34
	Completely Inelastic Collisions	37
	Elastic Collisions	38
8.4	Center of Mass	40
	Motion of the Center of Mass	42

8.5	Center-of-Mass Frame	44
9	Rotation of Rigid Bodies	47
9.1	Angular Kinematics	47
9.2	Energy in Rotational Motion	52
	Gravitational Potential Energy of Extended Bodies	55
9.3	Parallel Axis Theorem	56
9.4	Moment-of-Inertia Derivations	56
10	Dynamics of Rotational Motion	62
10.1	Torque	62
10.2	Torque and Angular Acceleration	63
10.3	Rotation About a Moving Axis	66
	Rolling without Slipping	66
10.4	Work and Power in Rotational Motion	68
10.5	Angular Momentum	69
	Angular Momentum on a Rigid Body	69
10.6	Conservation of Angular Momentum	70
11	Equilibrium and Elasticity	73
11.1	Conditions for Equilibrium	73
11.2	Center of Gravity	73
11.3	Rigid Body Equilibrium Problems	74
11.4	Stress, Strain, and Elastic Moduli	78
	Tensile and Compressive Stress and Strain	78
11.5	Elasticity and Plasticity	79
14	Periodic Motion	81
14.1	Describing Oscillation	81
14.2	Simple Harmonic Motion	81
	Relating Circular Motion to SHM	82
	Kinematics in SHM	84
14.3	Energy in SHM	85
14.4	Applications of SHM	85
	Vertical SHM	85
	Angular SHM	86
14.5	Simple Pendulum	86
14.6	Physical Pendulum	87

1 Units, Physical Quantities, and Vectors

1.1 SI Units

- **SI Units** refer to the system of units used by engineers and scientists across the world.
- For PHYS 161, these will be the most commonly used units in problems.

Application	Unit	Shorthand
Time	Seconds	s
Length	Meter	m
Mass	Kilogram	kg

- Additionally, many prefixes may be used in problems, necessitating conversions.

Length	Mass	Time
1 nanometer = 1 nm = 10^{-9} m	1 microgram = 1 μ g = 10^{-9} kg	1 nanosecond = 1 ns = 10^{-9} s
1 micrometer = 1 μ m = 10^{-6} m	1 milligram = 1 mg = 10^{-6} kg	1 microsecond = 1 μ s = 10^{-6} s
1 millimeter = 1 mm = 10^{-3} m	1 gram = 1 g = 10^{-3} kg	1 millisecond = 1 ms = 10^{-3} s

1.2 Unit Conversions

- In equations, quantities may only be added if they have the same dimension.
- When multiplying two quantities, the product's dimension is the product of the operands' dimensions.

$$(m/s)(s) = m$$

1.3 Significant Figures

- The **uncertainty** of a measurement is often indicated by the number of important digits, known as the **significant figures**.
 - All non-zero digits are significant figures.

$$525 \Rightarrow \text{three significant figures}$$

- Meanwhile, zero digits are only significant if they are between non-zero digits, or they are trailing non-zero digits when there is a decimal point.

40001 $\Rightarrow$ five significant figures

120.0 $\Rightarrow$ four significant figures

- When adding quantities, the significant figures of the sum are equal to the operand with the fewest significant figures.

$$25.7 + 13 = 38.7 \approx 39 \Rightarrow \text{two significant figures}$$

- When multiplying quantities, the same rule applies to the product.

$$(13.7)(20.95) = 287.015 \approx 287 \Rightarrow \text{three significant figures}$$

Important

In longer physics problems with numerous equations, using the actual values in calculations and only converting the final answer to the correct number of significant figures is best practice.

1.4 Vectors

- **Scalars** are quantities which only have *magnitude*.
 - An example of a scalar would be mass.
- **Vectors** are quantities with both *magnitude* and *direction*.
 - An example of a vector would be displacement.
 - Vector variables are denoted with an arrow or are bolded.
 - Their components are denoted with subscripts that indicate which direction they represent:

$$\mathbf{u} = \vec{u} = \langle u_x, u_y, u_z \rangle$$

- For this course, we often use **directional unit vectors** to describe direction:

$$\hat{i} = \langle 1, 0, 0 \rangle$$

$$\hat{j} = \langle 0, 1, 0 \rangle$$

$$\hat{k} = \langle 0, 0, 1 \rangle$$

$$\vec{u} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k}$$

- A vector's magnitude is often denoted by double or single bars surrounding it or by removing the arrow symbol. It can be calculated using the components of the vector.

$$\|\vec{u}\| = u = \sqrt{u_x^2 + u_y^2 + u_z^2}$$

- **Unit vectors** are vectors with a magnitude of 1 and use a caret rather than an arrow.

$$\hat{u} = \frac{\vec{u}}{\|\vec{u}\|}$$

- **Vector addition** is done by adding corresponding components, resulting in a sum vector.

$$\vec{u} + \vec{v} = (u_x + v_x)\hat{i} + (u_y + v_y)\hat{j} + (u_z + v_z)\hat{k}$$

- **Scaling a vector** is done by multiplying each component in the vector by the scalar.

$$c\vec{u} = cu_x\hat{i} + cu_y\hat{j} + cu_z\hat{k}$$

Vector Multiplication

- The **dot product** or **scalar product** involves multiplying corresponding components, then summing them up. There also exists an alternate definition using the angle between the operands.

$$\vec{u} \cdot \vec{v} = u_xv_x + u_yv_y + u_zv_z = \|\vec{u}\|\|\vec{v}\|\cos(\theta)$$

- The **cross product** or **vector product** is perpendicular to the plane formed by the operands, and its magnitude is calculated using the angle between the operands, albeit with sine. There also exists a determinant definition, generally done with a cofactor expansion down the first row.

$$\vec{u} \times \vec{v} = \|\vec{u}\|\|\vec{v}\|\sin(\theta)\hat{u}_\perp = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix}$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} u_y & u_z \\ v_y & v_z \end{vmatrix} \hat{i} - \begin{vmatrix} u_x & u_z \\ v_x & v_z \end{vmatrix} \hat{j} + \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} \hat{k}$$

$$\vec{u} \times \vec{v} = (u_yv_z - u_zv_y)\hat{i} - (u_xv_z - u_zv_x)\hat{j} + (u_xv_y - u_yv_x)\hat{k}$$

2 Motion Along a Straight Line

2.1 Displacement, Velocity, and Acceleration

- **Time** is measured in seconds, and usually refers to the time passed between two particular points in time.
 - Denoted t .
- **Displacement** and **distance** is measured in meters.
 - Denoted Δx and x , respectively, along the x -axis.
- **Velocity** is measured in meters per second.
 - Denoted v .
- **Acceleration** is measured in meters per second squared.
 - Denoted a .

$$\Delta x = x_2 - x_1$$

$$v_{\text{av}} = \frac{\Delta x}{\Delta t}$$

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t}$$

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

2.2 Motion with Constant Acceleration

Definition 2.2.1 (Kinematic Equations)

With constant acceleration, the following **kinematic equations** hold:

$$v = v_0 + a\Delta t$$

$$\Delta x = v_0\Delta t + \frac{1}{2}a\Delta t^2$$

$$v^2 = v_0^2 + 2a\Delta x$$

Remark

When these equations are used, the Δ is sometimes omitted for aesthetics. However, the t and x are still deltas in practice.

Derivation 2.2.2 (Displacement Kinematic Equation)

$$\begin{aligned}
 v &= v_0 + a\Delta t \\
 \int_{t_1}^{t_2} v \, dt &= \int_{t_1}^{t_2} (v_0 + a\Delta t) \, dt \\
 \int_{t_1}^{t_2} \frac{dx}{dt} \, dt &= \int_{t_1}^{t_2} (v_0 + a\Delta t) \, dt \\
 \int_{x_1}^{x_2} dx &= \int_{t_1}^{t_2} (v_0 + a\Delta t) \, dt \\
 \Delta x &= v_0\Delta t + \frac{1}{2}a\Delta t^2
 \end{aligned}$$

Derivation 2.2.3 (Velocity Squared Kinematic Equation)

$$\begin{aligned}
 v &= v_0 + a\Delta t \\
 \Delta t &= \frac{\Delta v}{a} \\
 \Delta x &= v_0\Delta t + \frac{1}{2}a\Delta t^2 \\
 \Delta x &= v_0\left(\frac{\Delta v}{a}\right) + \frac{1}{2}a\left(\frac{\Delta v}{a}\right)^2 \\
 \Delta x &= \frac{vv_0 - v_0^2}{a} + \frac{v^2 - 2vv_0 + v_0^2}{2a} \\
 \Delta x &= \frac{v^2 - v_0^2}{2a} \\
 v^2 - v_0^2 &= 2a\Delta x \\
 v^2 &= v_0^2 + 2a\Delta x
 \end{aligned}$$

3 Motion in Two or Three Dimensions

3.1 Vector Quantities

- Many of the definitions from one-dimensional motion hold true in two or three dimensions, albeit using vectors.

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$$

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$

$$a = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

3.2 Projectile Motion

- In **projectile motion**, an object in the air only experiences acceleration due to gravity.
- Because the same kinematic equations can be applied *per component* of the vectors, the following generalizations can be made (given that z is the vertical direction):

$$\begin{aligned} v_x &= v_{0x} \\ \Delta x &= v_{0x}\Delta t \end{aligned}$$

$$\begin{aligned} v_y &= v_{0y} \\ \Delta y &= v_{0y}\Delta t \end{aligned}$$

$$\begin{aligned} v_z &= v_{0z} - g\Delta t \\ \Delta z &= v_{0z}\Delta t - \frac{1}{2}g\Delta t^2 \\ v_z^2 &= v_{0z}^2 - 2g\Delta z \end{aligned}$$

Constant g is the acceleration due to Earth's gravity, 9.81 m/s^2 . Note that y is generally the vertical direction when dealing with two dimensions.

- A common application of these formulas is finding the *air time* of a projectile. Thus, sometimes the quadratic equation may be necessary to solve for the time.
 - In this case, either the negative time or the later time will be disregarded.

$$t = \frac{-v_{0y} \pm \sqrt{v_{0y}^2 + 4a\Delta y}}{2a}$$

- Now, we can derive a general formula for a projectile's **range** and **maximum height**.

Derivation 3.2.1 (Range Formula)

Let Δx equal R and let Δy equal 0.

$$v_y = v_{0y} - gt$$

$$gt = v_{0y} - v_y$$

$$t = \frac{v_0 \sin \theta - v \sin \theta}{g}$$

$$t = \frac{v_0 \sin \theta - (-v_0 \sin \theta)}{g}$$

$$t = \frac{2v_0 \sin \theta}{g}$$

Now, we can plug this value for t into a horizontal kinematic equation.

$$R = v_{0x}t$$

$$R = v_0 \cos \theta t$$

$$R = v_0 \cos \theta \frac{2v_0 \sin \theta}{g}$$

$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

Derivation 3.2.2 (Maximum Height Formula)

Let Δy equal $h_{\max}$ and $v_y = 0$.

$$0 = v_{0y}^2 - 2gh_{\max}$$

$$2gh_{\max} = v_{0y}^2$$

$$h_{\max} = \frac{v_{0y}^2}{2g}$$

$$h_{\max} = \frac{v_0^2 \sin^2 \theta}{2g}$$

Note

In later chapters, there will be other ways of finding the range or maximum height of a projectile, but these may still be useful depending on what is known in a problem.

⚠ Caution

Because the maximum height and range formulas aren't generally standard physics equations, they should be clearly derived before using them.

Example 3.2.3 (Fireworks Display)

During a fireworks display, a shell is shot into the air with an initial speed $v_0 = 70.0$ m/s at an angle of 75° above the horizontal. The fuse is timed to ignite the shell just as it reaches its highest point above the ground.

- 1) How much time has passed between the launch of the shell and the explosion?
- 2) What is the horizontal displacement of the shell before it explodes?

1) This is essentially the time needed for the firework to reach its max height. We can derive the maximum height formula (as seen above) first.

$$h_{\max} = \frac{(70.0 \text{ m/s})^2 \sin^2 75^\circ}{2(9.81 \text{ m/s}^2)} = 233 \text{ m}$$

Now, we can solve for time easily.

$$h_{\max} = v_{0y}t - \frac{1}{2}gt^2$$

$$h_{\max} = v_0 \sin(75^\circ)t - \frac{1}{2}gt^2$$

$$\frac{1}{2}gt^2 - v_0 \sin(75^\circ)t + h_{\max} = 0$$

$$t = 6.89 \text{ s}$$

2) We can calculate the firework's displacement using the time we just calculated.

$$\Delta x = v_{0x}t$$

$$\Delta x = v_0 \cos(75^\circ)t$$

$$\Delta x = (70 \text{ m/s}) \cos(75^\circ)(6.89 \text{ s})$$

$$\Delta x = 125 \text{ m}$$

3.3 Circular Motion

- Even if a particle's speed is constant, its velocity cannot be constant in circular motion because *its direction is constantly changing*.
- The direction of the velocity changes due to acceleration directed toward the center of its rotation, known as **centripetal acceleration**.
 - Denoted a_c or a_{rad} .

- Thus, when centripetal acceleration is the only acceleration vector acting on a particle, the particle undergoes **uniform circular motion** because it will always be perpendicular to the particle's velocity vector.
- An acceleration vector parallel to the particle's velocity vector is known as **tangential acceleration**.
 - Denoted a_t .
 - This is known as **nonuniform circular motion**, because the speed of the particle will also be changing.

$$a_t = \frac{dv}{dt}$$

- **Frequency** is the rate at which an object spins, measured in revolutions per second (rev/s) or sometimes just in hertz ($s^{-1} = \text{Hz}$).
 - Denoted f .
- **Period** is the time required to make one revolution, measured in seconds.
 - Denoted T .

$$T = \frac{1}{f}$$

Definition 3.3.1 (Circular Motion Equations)

Given radius of the circular motion R :

$$v = \frac{2\pi R}{T} = 2\pi Rf$$

$$a_c = \frac{v^2}{R} = \frac{4\pi^2 R}{T^2} = 4\pi^2 Rf^2$$

$$a_{\text{tot}} = \left\| \frac{d\vec{v}}{dt} \right\| = \sqrt{a_c^2 + a_t^2}$$

Note

In later chapters, we will often incorporate angular speed ω into these equations.

3.4 Relative Motion

- Given reference frames A , B , and C where AB means “ A in the reference frame of B ”

$$\vec{v}_{AB} + \vec{v}_{BC} = \vec{v}_{AC}$$

$$\vec{v}_{AB} = -\vec{v}_{BA}$$

- This is essentially vector addition.
- The **Law of Sines** and the **Law of Cosines** are useful formulas for relative motion problems whenever angles are needed.

Definition 3.4.1 (Law of Sines)

Given sides a , b , and c opposite of angles A , B , and C , respectively,

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

**Definition 3.4.2 (Law of Cosines)**

Given sides a , b , and c opposite of angles A , B , and C , respectively,

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

**Example 3.4.3 (Metro Trains)**

Two metro trains pass by each other on the red line. The northbound train is moving at a speed of 25.0 m/s and the southbound train is moving at a speed of 20.0 m/s. In the northbound train, Cecilia is walking toward the front of the train at 2.0 m/s, and in the southbound one, Jack is moving toward the back of the train also at 2.0 m/s. What is the magnitude of Jack's velocity relative to Cecilia?

We can start by finding both of their speeds relative to the Earth.

$$\vec{v}_{\text{Cecilia,Earth}} = \vec{v}_{\text{Cecilia,North}} + \vec{v}_{\text{North,Earth}}$$

$$\vec{v}_{\text{Cecilia,Earth}} = (2 \text{ m/s}) + (25 \text{ m/s}) = 27 \text{ m/s}$$

$$\vec{v}_{\text{Jack,Earth}} = \vec{v}_{\text{Jack,South}} + \vec{v}_{\text{South,Earth}}$$

$$\vec{v}_{\text{Jack,Earth}} = (2 \text{ m/s}) + (-20 \text{ m/s}) = -18 \text{ m/s}$$

Now, we can solve for Jack's velocity relative to Cecilia.

$$\vec{v}_{\text{Jack,Cecilia}} = \vec{v}_{\text{Jack,Earth}} + \vec{v}_{\text{Earth,Cecilia}}$$

$$\vec{v}_{\text{Jack,Cecilia}} = \vec{v}_{\text{Jack,Earth}} - \vec{v}_{\text{Cecilia,Earth}}$$

$$\vec{v}_{\text{Jack,Cecilia}} = (-18 \text{ m/s}) - (27 \text{ m/s})$$

$$\vec{v}_{\text{Jack,Cecilia}} = -45.0 \text{ m/s}$$

$$\boldsymbol{v}_{\text{Jack,Cecilia}} = \mathbf{45.0 \text{ m/s}}$$



4 Newton's Laws of Motion

4.1 Newton's First Law

Definition 4.1.1 (Newton's First Law)

An object subject to no external forces is at rest or moves with constant velocity if viewed from an inertial reference frame.

Inertial Reference Frame $\equiv$ Reference frame in which Newton's laws are valid

- An inertial reference frame must not be accelerating or rotating.

4.2 Newton's Second Law

- **Forces** are pushes or pulls on objects, measured in Newtons (N) or $(\text{kg} \cdot \text{m/s}^2)$.

Definition 4.2.1 (Newton's Second Law)

$$\Sigma \vec{F} = m\vec{a}$$

4.3 Newton's Third Law

Definition 4.3.1 (Newton's Third Law)

For every action there is an equal and opposite reaction.

Given objects A and B , where the subscript AB means “exerted from A on B ”:

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

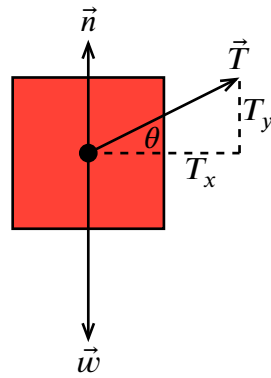
- For example, when two objects collide, they experience the same magnitude of force from the collision.

4.4 Free Body Diagrams

- **Free-body diagrams** show the forces acting on an object's center, including any relevant components.
- While the magnitude of the drawn forces isn't that important, scaling them relative to each other makes the direction of the net acceleration clearer.

⚠ Caution

When indicating forces, try to distinguish the system's acceleration from the force vectors. Acceleration is not a force, and the net acceleration is often from multiple forces.



Free-body diagram of a block being pulled at an angle θ by a rope. Note how the vertical components of the forces are all balanced, signifying that it is being dragged horizontally.

5 Applying Newton's Laws

5.1 Types of Forces

- **Fundamental forces** are the four natural interactions that govern how particles interact.
 - **Weight**
 - Electromagnetic
 - Weak nuclear
 - Strong nuclear
- **Non-fundamental forces** are forces whose interactions are derived from the fundamental forces.
 - **Spring**
 - **Contact** (tension, normal, friction, ...)
 - Microscopic
- The **weight** force is from gravity. On Earth, it is defined as follows:

$$\vec{w} = m\vec{g}$$

- However, there also exists a general definition for weight force, the **Universal Law of Gravitation** using the gravitational constant G :

$$G = 6.6743 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-1}$$

Definition 5.1.1 (Universal Law of Gravitation)

The force between any two objects is defined as follows, given that r is the *distance between the objects' centers*:

$$F_1 = F_2 = G \frac{m_1 m_2}{r^2}$$

- The **spring force** is from compressing a spring.
 - Springs have differing **spring constants** usually denoted as k , and they define the ratio between the force exerted by the spring and the distance compressed from the equilibrium position.

$$\vec{F}_{\text{spr}} = -k\vec{x}$$

- The **normal force** is exerted by a surface on an object.
 - Generally denoted $\vec{n}$.
- The **tension force** is the pulling force exerted equally throughout a rope, cable, etc.
 - Generally denoted $\vec{T}$.
- The **friction force** is exerted by a surface on an object and acts perpendicular to the normal force. It depends on the roughness of the surface (expressed by the coefficient of friction).
 - **Kinetic friction**, denoted $\vec{f}_k$ with coefficient of kinetic friction μ_k , acts on objects sliding across a surface.

$$\vec{f}_k = \mu_k \vec{n}$$

- **Static friction**, denoted $\vec{f}_s$ with coefficient of static friction μ_s , acts on still objects until the applied force exceeds the maximum static friction.

$$\vec{f}_s \leq \vec{f}_{s,\max} = \mu_s \vec{n}$$

5.2 Forces in Equilibrium

- Objects are in equilibrium when they are at rest or moving with constant velocity, meaning that the net force acting on those objects must be zero.

$$\Sigma \vec{F} = 0$$

- A common example of forces in equilibrium is an object sitting on a horizontal surface. The net vertical force is zero because the magnitude of the normal force and the weight force are equal.

5.3 Dynamics of Particles

- Using Newton's second law, we can set all forces (or components of forces) equal to mass times acceleration and solve for unknowns accordingly.
- Signs can indicate the direction, but you have to be consistent.
 - In **pulley problems**, direction should be based on the direction of the pulley's cord.
 - In **incline problems**, the x -axis should be parallel to the incline and the y -axis should be perpendicular to the incline.
 - Consequently, when given the angle of the incline, sine is used to find the x -component of weight and cosine is used to find the y -component of weight because weight will not be perpendicular to the incline.
- Sometimes, normal force can be referred to as **apparent weight** because the magnitude may be less than or greater than the opposing weight force.
 - For instance, you feel *heavier* when you are in an elevator moving up because your normal force is greater than your weight force and vice versa.

$$n = m(g + a_y)$$

Example 5.3.1 (Table Pulley)

Block B with mass $m_B = 5.0$ kg rests on block A, with mass $m_A = 8.0$ kg, which is on a horizontal frictionless tabletop. The coefficient of static friction between block A and block B is $\mu_s = 0.75$. A string attached to block A passes over a massless, frictionless pulley at the edge of the tabletop, and block C is suspended from the other end of the string and is released from rest.

- 1) Write Newton's second law for each of the three blocks in both directions.
- 2) Find the largest mass that block C can have so that blocks A and B slide together when the system is released from rest.

1) Along the direction of the pulley, the vertical forces acting on block A and B are irrelevant.

$$\Sigma F_A = T - f_s \quad \Sigma F_B = f_s \quad \Sigma F_C = m_C g - T$$

2) Now, blocks A and B will slip once the maximum static friction is exceeded, so we can use their net force equations to find the acceleration of the system when m_C is maximized.

$$\begin{aligned} m_B a &= \mu_s m_B g \\ a &= \mu_s g \\ a &= (0.75)(9.81 \text{ m/s}^2) = 7.36 \text{ m/s}^2 \end{aligned}$$

Using this, we can find an expression for T .

$$\begin{aligned} T - f_s + f_s &= (m_A + m_B) a \\ T &= (m_A + m_B) a \end{aligned}$$

With both acceleration and tension found, we can now solve for the maximum mass of C.

$$\begin{aligned} m_C a &= m_C g - T \\ m_C a &= m_C g - (m_A + m_B) a \\ m_C a - m_C g &= -(m_A + m_B) a \\ m_C &= -\frac{(m_A + m_B) a}{a - g} \\ m_C &= -\frac{(8 \text{ kg} + 5 \text{ kg})(7.36 \text{ m/s}^2)}{7.36 \text{ m/s}^2 - 9.81 \text{ m/s}^2} \\ m_C &= \mathbf{39 \text{ kg}} \end{aligned}$$

5.4 Dynamics of Circular Motion

- Based on Newton's Second Law, circular motion can result from many different forces.

$$\Sigma F = ma_c = \frac{mv^2}{R}$$

⚠ Caution

Denoting a singular “centrifugal force” is generally not accurate.

- For example, if an object is moving vertically with uniform circular motion, the weight force points toward the center of rotation at the top, but away from the center of rotation at the bottom.
- Additionally, direction should be considered relative to the center of rotation—generally, the positive direction is toward the center.
- Additionally, problems involving **circular motion of vehicles** have less obvious forces contributing to the circular motion.
 - Along a curved road with friction, *static friction is what maintains the vehicle's circular motion* because at any given moment, the tires of the car are stationary relative to the pavement.
 - Along a banked curve without friction (sort of like an inclined plane from the inner curve to the outer curve), *the x-component of the normal force is what maintains the vehicle's circular motion*.

Example 5.4.1 (Rounding a Flat Curve)

A sports car is rounding a flat, unbanked curve with radius R . If the coefficient of static friction between the tires is μ_s , what is the maximum speed $v_{\max}$ the driver can complete the curve without sliding off?

Here, the only force contributing to the circular motion is static friction, so we can just use the given formula for maximum static friction to find $v_{\max}$.

$$\begin{aligned} ma_c &= f_s \\ \frac{mv_{\max}^2}{R} &= \mu_s mg \\ v_{\max} &= \sqrt{R\mu_s g} \end{aligned}$$

Note that the maximum speed needed is not based on the mass of the vehicle at all.

Example 5.4.2 (Rounding a Banked Curve)

You are planning to create a road banked at the perfect angle to allow cars to complete turns without friction. At what angle α should the road be banked to allow for this?

We can imagine this from a straight back perspective of the car. While similar to an inclined plane problem, we are evaluating components parallel to the horizontal rather than the incline. As a result, the only inward force is the x-component of the normal force because the curve is banked from the inside-out.

Now, because the normal force and its y-component are perpendicular to the respective sides forming the angle α , *the angle between them is also equal to α* . Thus, the x-component of the normal force is the component opposite of α while the y-component of the normal force is the component adjacent of α .

$$\begin{aligned}n_y &= mg \\n \cos \alpha &= mg \\n &= \frac{mg}{\cos \alpha}\end{aligned}$$

Using this value of n , we can now solve for the angle.

$$\begin{aligned}n_x &= \frac{mv^2}{R} \\n \sin \alpha &= \frac{mv^2}{R} \\mg \frac{\sin \alpha}{\cos \alpha} &= \frac{mv^2}{R} \\mg \tan \alpha &= \frac{mv^2}{R} \\\tan \alpha &= \frac{v^2}{Rg} \\\alpha &= \arctan\left(\frac{v^2}{Rg}\right)\end{aligned}$$

◇

6 Work and Kinetic Energy

6.1 Work

- A force does **work** on a body if the body undergoes a displacement.
- The units of work are joules (J) or newton-meters.

Definition 6.1.1 (Work)

Given force $\vec{F}$ and displacement $\vec{s}$, work is defined as follows:

$$W = \vec{F} \cdot \vec{s} = F s \cos(\theta)$$

- Because work is defined by a dot product, it is a scalar quantity, although it can still be negative.
- Thus, it follows that a force does **negative work** when applied opposite of the direction of $\vec{s}$ and does **zero work** when applied perpendicular to the direction of $\vec{s}$.
- Additionally, given this definition, there are multiple ways to define **net work**.

$$W_{\text{net}} = \Sigma \vec{F} \cdot \vec{s} = \Sigma W$$

6.2 Kinetic Energy

- **Energy** is the capacity to do work, and also uses joules.
- **Kinetic energy** is the energy of a body from its mass and speed.

Definition 6.2.1 (Kinetic Energy)

Given mass m and speed v , kinetic energy is defined as follows:

$$K = \frac{1}{2}mv^2$$

- Because kinetic energy is based on speed, it is either positive or zero.
- Additionally, work may be expressed as the **change in kinetic energy**.

Theorem 6.2.2 (Work-Energy Theorem)

Given work W , kinetic energy K , and displacement s :

$$F_{\parallel} = ma = m \frac{v_2^2 - v_1^2}{2s}$$

$$F_{\parallel}s = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \Delta K$$

Important

These formulas only hold when the force remains constant during the displacement.

Example 6.2.3 (Final Velocity of a Sled)

You push a 30.0 kg sled with constant horizontal force $F = 500$ N for a horizontal distance $d = 20.0$ m. The sled's initial velocity is 10.0 m/s. What is the final velocity of the sled?

1) Because we can calculate the work done and we are given the initial velocity, we can use the work-energy theorem to derive an expression for the final velocity.

$$W = F_{\parallel} d$$

$$W = (500 \text{ N})(20 \text{ m})$$

$$W = 10000 \text{ J}$$

$$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = 10000 \text{ J}$$

$$\frac{1}{2}mv_2^2 = 10000 \text{ J} + \frac{1}{2}mv_1^2$$

$$v_2 = \sqrt{\frac{20000 \text{ J}}{m} + v_1^2}$$

$$v_2 = \sqrt{\frac{20000 \text{ J}}{30.0 \text{ kg}} + (10 \text{ m/s})^2}$$

$$v_2 = 27.7 \text{ m/s}$$

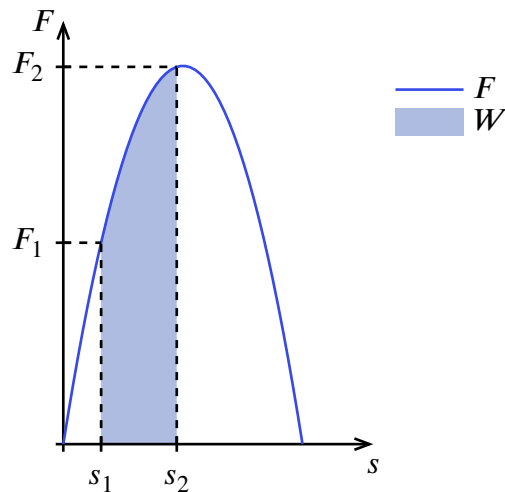
6.3 Work and Energy with Varying Force

- Given a constant net force F_{av} , we can define work as follows:

$$W = F_{\text{av}} \Delta s$$

- Thus, we can use an integral to calculate work done by a net force.

$$W = \int_{s_1}^{s_2} F \, ds$$



- In more than one dimension:

$$W = \int_{r_1}^{r_2} \vec{F} \cdot d\vec{r}$$

- This is also the definition for work done by a spring because it varies depending on the displacement.

Definition 6.3.1 (Work Done by a Spring)

$$W_s = \int_{x_1}^{x_2} (-kx) dx$$

$$W_s = -\frac{1}{2}k\Delta x^2$$



6.4 Power

- **Power** is the rate at which work is done, measured in watts (W) or joules per second (J/s).
 - Denoted P .
 - Power is also sometimes given in horsepower (hp). The conversion rate is as follows:

$$1 \text{ hp} = 745.7 \text{ W}$$

- Thus, average power is defined as follows:

$$P_{\text{av}} = \frac{\Delta W}{\Delta t}$$

Definition 6.4.1 (Power)

Given work W , velocity $\vec{v}$, and force $\vec{F}$, power is defined as follows:

$$P = \frac{dW}{dt} = \vec{F} \cdot \vec{v} = Fv \cos \theta$$



7 Potential Energy and Energy Conservation

- **Potential energy** is energy based on position, and may be *converted* into other forms of energy depending on the change in position.
 - For example, a fruit hanging from a tall tree has the *potential* to reach a higher speed (and therefore kinetic energy) before it hits the ground than a fruit hanging from a smaller tree.

7.1 Gravitational Potential Energy

- **Gravitational potential energy** is potential energy dependent on an object's mass and height above the ground.
 - Denoted U_g .

Definition 7.1.1 (Gravitational Potential Energy)

Given object's mass m , acceleration due to gravity g , and height above the ground h , gravitational potential energy is defined as follows:

$$U_g = mgh$$

- Notice how an object's gravitational potential energy *decreases* as its height above the ground decreases, even though the work done by the weight force on a falling object is *positive*.
- Thus, the work done by gravity is equal to the negative change in gravitational potential energy.

$$W_g = -\Delta U_g$$

7.2 Elastic Potential Energy

- **Elastic potential energy** is potential energy stored in deformable objects such as springs when they are stretched or compressed from equilibrium.
 - Denoted U_s .

Definition 7.2.1 (Elastic Potential Energy)

Given spring constant k and elongation of object x , elastic potential energy is defined as follows:

$$U_s = \frac{1}{2}kx^2$$

- Similarly to work done by gravity, because an object's elastic potential energy increases the more it is displaced from equilibrium, the work done by the spring force is equal to the negative change in elastic potential energy.

$$W_s = -\Delta U_s$$

7.3 Conservative and Nonconservative forces

- **Mechanical energy** is energy that is either kinetic or potential.
- **Conservative forces** do work independent of the path, meaning that they only depend on the endpoints of the displacement.
 - The work done by conservative forces is reversible because they convert between forms of mechanical energy.
- **Nonconservative forces** do work dependent on the path.
 - The work done by nonconservative forces is irreversible because they convert forms of mechanical energy to other forms of energy.

The Law of Conservation of Energy

- Thus, as long as the net work done by nonconservative forces in a system is zero, the change in total mechanical energy is zero.
- As a result, we can equate the total mechanical energy at different points in a system.
- Thus, work done by nonconservative forces should be subtracted.

Definition 7.3.1 (Law of Conservation of Energy)

Given kinetic energy K , potential energy U , and work from nonconservative forces W_{nc} :

$$\Delta K + \Delta U - W_{nc} = 0$$

- For conservation of energy problems, the following form is often used:

$$K_1 + U_1 + W_{nc} = K_2 + U_2$$

- Nonconservative forces may also be represented using **internal energy** U_{int} such that the internal energy changes when external forces do work on the system.

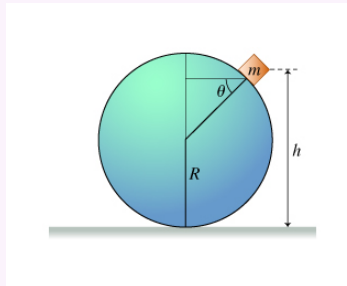
$$\Delta U_{int} = -W_{nc}$$

Note

As positive work is done on an object, they tend to displace to positions of lower potential energy.

Example 7.3.2 (Block on a Globe)

A small block of mass m sits on top of a globe with radius R . The small block begins sliding from the top of the globe with negligible speed. The block leaves the surface of the globe when it reaches a height h_{crit} above the ground. Derive a formula for v_{crit} , the speed of the block the moment it falls off the globe, in terms of R , h_{crit} , and g .



The first step is to identify two points in the system that would include the desired terms. One of them should be when the block is falling off because it is at height h_{crit} . The other point should be at the top of the globe because the height above the ground is just the diameter of the globe.

$$\begin{aligned}\frac{1}{2}mv_{\text{crit}}^2 + mgh_{\text{crit}} &= mg(2R) \\ \frac{1}{2}mv_{\text{crit}}^2 &= mg(2R) - mgh_{\text{crit}} \\ v_{\text{crit}}^2 &= 4gR - 2gh_{\text{crit}} \\ v_{\text{crit}} &= \sqrt{4gR - 2gh_{\text{crit}}}\end{aligned}$$

Example 7.3.3 (Throwing a Baseball)

The maximum horizontal distance you can throw a 0.145 kg baseball is 25 m at an angle of $\alpha = 45^\circ$ above the horizontal.

- 1) Ignoring air resistance and the small distance the baseball is above the ground when it is thrown, what is the kinetic energy of the baseball the moment it is thrown?
- 2) At the ball's maximum height, what fraction of its total mechanical energy is kinetic energy?

1) Because we are given the range of the baseball but not the initial velocity, we can use our previously derived range formula to find the initial velocity.

Continued on next page

Example (Throwing a Baseball *continued*)

$$R = \frac{v_0^2 \sin(2\alpha)}{g}$$

$$v_0^2 = \frac{Rg}{\sin(2\alpha)}$$

$$v_0 = \sqrt{\frac{Rg}{\sin(2\alpha)}} = \sqrt{\frac{(25 \text{ m})(9.81 \text{ m/s}^2)}{\sin 90^\circ}}$$

$$v_0 = 15.7 \text{ m/s}$$

Now we can plug this back into the kinetic energy formula.

$$K_0 = \frac{1}{2}mv_0^2$$

$$K_0 = \frac{1}{2}(0.145 \text{ kg})(15.7 \text{ m/s})^2$$

$$\mathbf{K_0 = 17.8 J}$$

2) Because we know that the vertical component of the velocity vector is zero at the highest point in its trajectory, the velocity at the maximum height is just the horizontal component.

$$K_1 = \frac{1}{2}mv_1^2$$

$$K_1 = \frac{1}{2}mv_0^2 \cos^2(45^\circ)$$

$$K_1 = \frac{1}{2}mv_0^2 \left(\frac{1}{2}\right)$$

$$K_1 = \frac{1}{2}K_0 = \frac{1}{2}E_{\text{tot}}$$

$$\frac{K_1}{E_{\text{tot}}} = \frac{1}{2}$$



Example 7.3.4 (Rough Patch)

A small block is sent through point A with a speed of 7 m/s. Point A is at height $h_1 = 6$ m. The block then slides down to point B, which is at a height of zero. After that, the block slides up to point C at a height of $h_2 = 2$ m. Past point C is a rough patch of length $L = 12$ m with a coefficient of kinetic friction $\mu_k = 0.7$.

If point D is just beyond the rough section, will the block be able to reach it? If not, how far through the rough patch can it travel?

The total mechanical energy at points A, B, and C are all equal. We know the block can reach point D if the friction requires a longer distance than 12 m for the kinetic energy by point C to be totally dissipated.

We can start by finding the velocity at point C because the gravitational potential energy of the block is constant between point C and point D.

$$\begin{aligned}
 E_A &= E_C \\
 \frac{1}{2}mv_A^2 + mgh_1 &= \frac{1}{2}mv_C^2 + mgh_2 \\
 \frac{1}{2}v_A^2 + gh_1 &= \frac{1}{2}v_C^2 + gh_2 \\
 \frac{1}{2}v_A^2 + g(h_1 - h_2) &= \frac{1}{2}v_C^2 \\
 v_A^2 + 2g(h_1 - h_2) &= v_C^2 \\
 v_C^2 &= (7 \text{ m/s})^2 + 2(9.81 \text{ m/s}^2)(6 \text{ m} - 2 \text{ m}) \\
 v_C^2 &= 127.48 \text{ m}^2/\text{s}^2
 \end{aligned}$$

Now, we can equate the kinetic energy at point C to the work done by friction and solve for the distance needed to fully dissipate it.

$$\begin{aligned}
 K_C &= W_f \\
 \frac{1}{2}mv_C^2 &= \mu_k mgd \\
 d &= \frac{v_C^2}{2\mu_k g} \\
 d &= \frac{127.48 \text{ m}^2/\text{s}^2}{2(0.7)(9.81 \text{ m/s}^2)} \\
 d &= \frac{127.48 \text{ m}^2/\text{s}^2}{2(0.7)(9.81 \text{ m/s}^2)} \\
 d &= 9.28 \text{ m}
 \end{aligned}$$

Thus, the block cannot reach point D, and will instead travel 9.28 m across the rough patch.

◇

7.4 Force and Potential Energy

- Recall that the work done by a conservative force equals the negative change in potential energy.

$$W = -\Delta U$$

- Thus, given force $\vec{F}$ and potential energy U :

$$\vec{F} = -\vec{\nabla}U = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k}$$

The “ ∂ ” symbol refers to taking a derivative with respect to one variable, holding everything else constant, known as a *partial derivative*. The nabla vector is known as the gradient vector, and is a vector consisting of all partial derivatives.

Note

This is a multivariable calculus concept.

Example 7.4.1 (Object’s Potential Energy Function)

A small object with mass $m = 0.0900$ kg moves along the $+x$ -axis. The only force applied to the object is a conservative force with potential energy function $U(x) = -\alpha x^2 + \beta x^3$ where

$\alpha = 9.50$ J/m² and $\beta = 0.300$ J/m³. The object is released from rest at negligible x .

- When the object is at $x = 4.00$ m, what is its speed?
- When the object is at $x = 4.00$ m, what is the magnitude of its acceleration?
- What is the maximum value of x reached by the object during its motion?

1) Because we know the object starts from rest, the work done by the conservative force must be equal to the kinetic energy. Thus, we can equate kinetic energy to the work done, or the negative potential energy function (because we start from a negligible x) and solve for the velocity.

$$K = W$$

$$\frac{1}{2}mv^2 = -U(4)$$

$$v = \sqrt{-\frac{2U(4)}{m}}$$

$$v = \sqrt{\frac{2(9.5 \text{ J/m}^2)(4 \text{ m})^2}{0.09 \text{ kg}} - \frac{2(0.3 \text{ J/m}^3)(4 \text{ m})^3}{0.09 \text{ kg}}}$$

$$v = 54.3 \text{ m/s}$$

Continued on next page

Example (Object's Potential Energy Function *continued*)

2) We can start by finding the negative derivative to get the the force function, then use the force function to get the acceleration function.

$$F = -\frac{dU}{dx}$$

$$F = 2\alpha x - 3\beta x^2 = ma$$

$$a = \frac{2\alpha x}{m} - \frac{3\beta x^2}{m}$$

$$a(4) = \frac{2(9.5 \text{ J/m}^2)(4 \text{ m})}{0.09 \text{ kg}} - \frac{3(0.3 \text{ J/m}^3)(4 \text{ m})^2}{0.09 \text{ kg}}$$

$$a(4) = \mathbf{684 \text{ m/s}^2}$$

3) Similarly to our logic for part one, we can use the fact that $K = W = -U(x)$. By finding the zeros of $-U(x)$, we find that $x_{\max}$ is whenever the kinetic energy is zero again: $x_{\max} = \mathbf{31.7 \text{ m}}$.

◇

8 Momentum, Impulse, and Collisions

8.1 Momentum and Impulse

- **Momentum** is an object's *quantity of motion*, measured in $\text{kg} \cdot \text{m/s}$ or $\text{N} \cdot \text{s}$
 - Denoted p .

Definition 8.1.1 (Momentum)

Given mass m and velocity $\vec{v}$, momentum is defined as follows:

$$\vec{p} = m\vec{v}$$

- Thus, momentum is a vector quantity.
- We can also redefine Newton's second law in terms of momentum:

$$\Sigma \vec{F} = m \left(\frac{d\vec{v}}{dt} \right) = \frac{d}{dt}(m\vec{v}) = \frac{d\vec{p}}{dt}$$

- An object's change in momentum is known as **impulse**.
 - Denoted J .

Theorem 8.1.2 (Impulse-Momentum Theorem)

Given impulse $\vec{J}$ and momentum $\vec{p}$:

$$\vec{J} = \Delta \vec{p}$$

Derivation 8.1.3 (Relating Impulse to Force)

Given net force $\Sigma \vec{F}$, acceleration $\vec{a}$, and mass m :

$$\Sigma \vec{F} = m\vec{a}$$

$$\Sigma \vec{F} = m \frac{d\vec{v}}{dt}$$

$$\Sigma \vec{F} dt = m d\vec{v}$$

$$m d\vec{v} = \Sigma \vec{F} dt$$

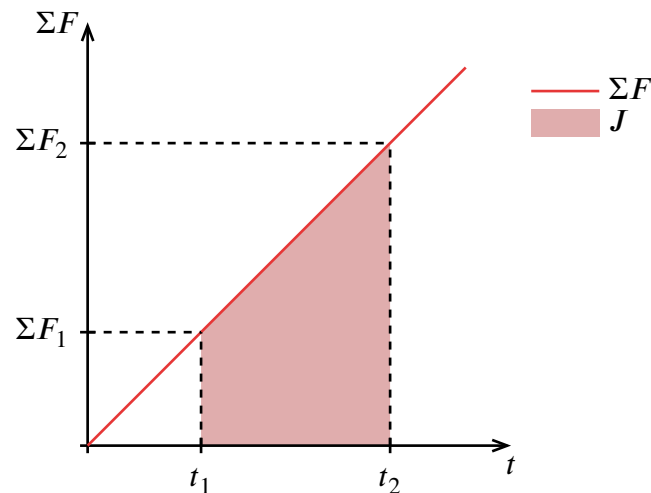
$$m \int_{v_1}^{v_2} d\vec{v} = \int_{t_1}^{t_2} \Sigma \vec{F} dt$$

$$m(\vec{v}_2 - \vec{v}_1) = \int_{t_1}^{t_2} \Sigma \vec{F} dt$$

$$\Delta p = \vec{J} = \int_{t_1}^{t_2} \Sigma \vec{F} dt$$

$$\vec{J} = \Sigma \vec{F}_{\text{av}} \Delta t$$

- It follows that impulse is the area under a force versus time graph.



Example 8.1.4 (Ball Rebounding)

You throw a 0.40 kg ball against a brick wall with a horizontal speed of 30 m/s. After colliding with the wall, the ball rebounds with a horizontal speed of 20 m/s.

- Find the impulse of the net external force on the ball during the impact.
- If the ball is in contact with the wall for 0.01 s, find the average horizontal force that the wall exerts on the ball during the impact.

1) Because we are given an initial and final velocity of the ball, we can apply the impulse-momentum theorem.

$$J_x = \Delta p_x$$

$$J_x = mv_{2x} - mv_{1x}$$

$$J_x = m(v_{2x} - v_{1x})$$

$$J_x = (0.4 \text{ kg})((-20 \text{ m/s}) - (30 \text{ m/s}))$$

$$J_x = -20 \text{ N} \cdot \text{s}$$

2) Now that we know the impulse of the net external force, the average force can be calculated using the fact that the impulse is the area under a force versus time graph.

$$J_x = \Sigma F_{\text{av-x}} \Delta t$$

$$\Sigma F_{\text{av-x}} \Delta t = J_x$$

$$\Sigma F_{\text{av-x}} = \frac{J_x}{\Delta t}$$

$$\Sigma F_{\text{av-x}} = \frac{20 \text{ N} \cdot \text{s}}{0.01 \text{ s}}$$

$$\Sigma F_{\text{av-x}} = 2000 \text{ N}$$

8.2 Conservation of Momentum

- **Internal forces** are the forces exerted by a system's particles on each other.
- **External forces** are the forces exerted by particles outside the system on the system's particles.
- **Isolated systems** are systems with no external forces.
- The **total momentum** of a system is the sum of the momenta of the system's particles.
 - Denoted $\vec{P}$.

$$\vec{P} = \vec{p}_A + \vec{p}_B + \cdots = m\vec{v}_A + m\vec{v}_B + \cdots$$

Definition 8.2.1 (Conservation of Momentum)

We say that a system's total momentum is *conserved* if the system's net external force is zero.

- Remember that momentum is a vector quantity, so apply its conservation per component in problems with more than one dimension.
- Note that by Newton's third law, a system's internal forces do not impact the system's total momentum.

Example 8.2.2 (Gun Recoil)

A marksman holds a 3.00 kg rifle loosely, allowing it to recoil. He fires a 5.00 g bullet whose horizontal velocity relative to the ground is 300 m/s.

- 1) What is the rifle's recoil velocity?
- 2) What are the final momentum and kinetic energy of the bullet and rifle?

1) During the shot, the external forces exerted by the marksman are negligible. Thus, the system's total momentum is conserved. Now, it follows that the system's total momentum is initially zero, so it should be zero after the shot. Since we are given everything but the rifle's recoil velocity, we can simply isolate it by equating the momenta to zero.

$$\begin{aligned} p_R + p_B &= 0 \\ p_R &= -p_B \\ m_R v_R &= -m_B v_B \\ v_R &= -\frac{m_B v_B}{m_R} \\ v_R &= -\frac{\left(5 \text{ g} \cdot \frac{0.001 \text{ kg}}{1 \text{ g}}\right)(300 \text{ m/s})}{3 \text{ kg}} \\ v_R &= -0.500 \text{ m/s} \end{aligned}$$

Continued on next page

Example (Gun Recoil *continued*)

2) Now that we have the recoil velocity of the rifle, we can simply plug our values into the momentum and kinetic energy formulas.

$$\begin{array}{llll}
 p_R = m_R v_R & K_R = \frac{1}{2} m_R v_R^2 & p_B = m_B v_B & K_B = \frac{1}{2} m_B v_B^2 \\
 p_R = -1.50 \text{ kg} \cdot \text{m/s} & K_R = 0.375 \text{ J} & p_B = 1.5 \text{ kg} \cdot \text{m/s} & K_B = 225 \text{ J}
 \end{array}$$

◇

8.3 Momentum Conservation and Collisions

- Collisions are strong interactions between objects over a relatively short time interval.
 - Momentum is conserved in collisions, but kinetic energy is not always conserved.
- **Elastic collisions** occur if the forces between the colliding objects are conservative.
 - Thus, the total kinetic energy is conserved.

$$\frac{1}{2} m_A v_{1A}^2 + \frac{1}{2} m_B v_{2A}^2 = \frac{1}{2} m_A v_{2A}^2 + \frac{1}{2} m_B v_{2B}^2$$

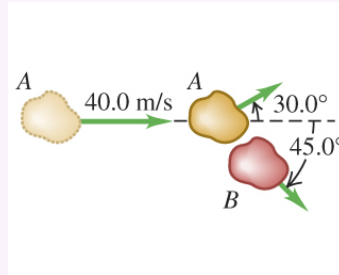
- **Inelastic collisions** occur if there are nonconservative forces between the colliding objects.
- **Completely inelastic collisions** are when colliding objects stick together and move as one object.
 - This leads to a unique formula for conservation of momentum in completely inelastic collisions:

$$m_A \vec{v}_A + m_B \vec{v}_B = (m_A + m_B) \vec{v}_{AB}$$

- In inelastic and completely inelastic collisions, kinetic energy is not conserved.

Example 8.3.1 (Asteroid Deflection)

Two asteroids of equal mass in the asteroid belt between Mars and Jupiter collide. Asteroid A initially travels with $v_{1A} = 40.0$ m/s with respect to an inertial frame where asteroid B is at rest. After the collision, asteroid A is deflected 30.0° from its original direction, while asteroid B is deflected 45.0° from asteroid A's original direction.



- 1) Find the speed of asteroid A after the collision.
- 2) Find the speed of asteroid B after the collision.

1) First, we want to find the velocity of asteroid B in terms of asteroid A. We can use the conservation of momentum in the direction perpendicular to asteroid A's initial trajectory to find it because the total momentum in this direction is zero.

$$\begin{aligned}
 0 &= mv_{2Ay} + mv_{2By} \\
 0 &= mv_{2A} \sin(30^\circ) + mv_{2B} \sin(45^\circ) \\
 -mv_{2B} \sin(45^\circ) &= mv_{2A} \sin(30^\circ) \\
 -v_{2B} &= v_{2A} \frac{\sin(30^\circ)}{\sin(45^\circ)} \\
 |v_{2B}| &= v_{2A} \frac{\sin(30^\circ)}{\sin(45^\circ)}
 \end{aligned}$$

Now, the only unknown for conservation of momentum in the x -direction is the final velocity of asteroid A in the x -direction, which can we express in terms of the final speed of asteroid A using trigonometry.

$$\begin{aligned}
 mv_{1Ax} &= mv_{2Ax} + mv_{2Bx} \\
 v_{1Ax} &= v_{2Ax} + v_{2Bx} \\
 v_{1Ax} &= v_{2A} \cos(30^\circ) + v_{2B} \cos(45^\circ) \\
 v_{1Ax} &= v_{2A} \cos(30^\circ) + v_{2A} \frac{\sin(30^\circ)}{\sin(45^\circ)} \cos(45^\circ) \\
 v_{1Ax} &= \left(\cos(30^\circ) + \frac{\sin(30^\circ)}{\sin(45^\circ)} \cos(45^\circ) \right) v_{2A}
 \end{aligned}$$

Continued on next page



Example (Asteroid Deflection *continued*)

$$v_{2A} = \frac{v_{1Ax}}{\cos(30^\circ) + \frac{\sin(30^\circ)}{\sin(45^\circ)} \cos(45^\circ)}$$

$$v_{2A} = \frac{40 \text{ m/s}}{\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2}}$$

$$v_{2A} = \mathbf{29.3 \text{ m/s}}$$

2) Because we just solved for the final speed of asteroid A, the final speed of asteroid B is easy to solve for using conservation of momentum.

$$v_{1Ax} = v_{2Ax} + v_{2Bx}$$

$$v_{2Bx} = v_{1Ax} - v_{2Ax}$$

$$v_{2B} \cos(45^\circ) = v_{1Ax} - v_{2A} \cos(30^\circ)$$

$$v_{2B} = \frac{v_{1Ax} - v_{2A} \cos(30^\circ)}{\cos(45^\circ)}$$

$$v_{2B} = \frac{40 \text{ m/s} - (29.3 \text{ m/s}) \frac{\sqrt{3}}{2}}{\frac{\sqrt{2}}{2}}$$

$$v_{2B} = \mathbf{20.7 \text{ m/s}}$$

◇

Completely Inelastic Collisions

Derivation 8.3.2 (Loss of Kinetic Energy in a Completely Inelastic Collision)

In order to establish why a system's kinetic energy decreases in completely inelastic collisions, we can first find the velocity after the inelastic collision using conservation of momentum. To simplify it further, we can say that object B is initially at rest.

$$(m_A + m_B)\vec{v}_{AB} = m_A\vec{v}_A + m_B\vec{v}_B$$

$$(m_A + m_B)\vec{v}_{AB} = m_A\vec{v}_A$$

$$\vec{v}_{AB} = \frac{m_A\vec{v}_A}{m_A + m_B}$$

Now, we can use their magnitudes to find initial and final kinetic energies of the system in terms of object A's original velocity and find the ratio between them.

$$K_1 = \frac{1}{2}m_A v_A^2$$

$$K_2 = \frac{1}{2}(m_A + m_B)v_{AB}^2$$

$$= \frac{1}{2}(m_A + m_B)\left(\frac{m_A v_A}{m_A + m_B}\right)^2$$

$$= \frac{m_A^2 v_A^2}{2(m_A + m_B)}$$

$$\therefore \frac{K_2}{K_1} = \frac{m_A}{m_A + m_B}$$

Now, it follows that this ratio is always less than one, showing that kinetic energy is always lost in completely inelastic collisions. This holds true even if object B is not originally at rest.

Caution

Remember that a collision that dissipates kinetic energy is not necessarily completely inelastic. Inelastic collisions can dissipate energy through other means such as deformations.

Example 8.3.3 (Ice Collision)

A 5.00 kg block of ice is sliding at 12.0 m/s across the floor of an icy valley when it collides and sticks to another 5.00 kg block of ice that is initially at rest. After the collision, how high above the valley floor will the combined chunks go?

Because the collision is inelastic, we first need to find the velocity of the combined blocks using conservation of momentum.

$$\begin{aligned}p_0 &= p_f \\mv_0 &= 2mv \\v &= \frac{1}{2}v_0\end{aligned}$$

Now, energy is conserved after the collision, and the kinetic energy is zero when the combined blocks reach their maximum height, allowing us to solve for it.

$$\begin{aligned}E_1 &= E_2 \\ \frac{1}{2}2mv^2 &= 2mgh \\ 2mgh &= mv^2 \\ h &= \frac{v^2}{2g} \\ h &= \frac{(0.5v_0)^2}{2g} \\ h &= \frac{(0.5 \cdot 12 \text{ m/s})^2}{2 \cdot 9.81 \text{ m/s}^2} \\ h &= \mathbf{1.83 \text{ m}}\end{aligned}$$

Elastic Collisions

- Commonly, elastic collision problems start with one object at rest before the collision.
- Thus, we can establish a relationship between the initial velocity of the moving object and the final velocity of the object initially at rest.

Derivation 8.3.4 (Final Velocities in an Elastic Collision)

To establish a relationship between the initial velocity of the moving object and the final velocity of the object initially at rest, we can apply the conservation of energy and momentum.

Continued on next page

Derivation (Final Velocities in an Elastic Collision *continued*)

$$\begin{aligned}\frac{1}{2}m_A v_{1A}^2 &= \frac{1}{2}m_A v_{2A}^2 + \frac{1}{2}m_B v_B^2 & m_A v_{1A} &= m_A v_{2A} + m_B v_B \\ \frac{1}{2}m_A (v_{1A}^2 - v_{2A}^2) &= \frac{1}{2}m_B v_B^2 & m_A (v_{1A} - v_{2A}) &= m_B v_B\end{aligned}$$

Now, by dividing the kinetic energy equation by the momentum equation, we can derive the final velocity of object B in terms of object A's velocities.

$$\begin{aligned}\frac{\frac{1}{2}m_A (v_{1A}^2 - v_{2A}^2)}{m_A (v_{1A} - v_{2A})} &= \frac{\frac{1}{2}m_B v_B^2}{m_B v_B} \\ \frac{1}{2}v_{1A} + \frac{1}{2}v_{2A} &= \frac{1}{2}v_B \\ v_B &= v_{1A} + v_{2A}\end{aligned}$$

Using this definition, we can define the final velocity of object A in terms of object A's initial velocity using the momentum equation.

$$\begin{aligned}m_A v_{1A} &= m_A v_{2A} + m_B (v_{1A} + v_{2A}) \\ m_A v_{1A} &= m_A v_{2A} + m_B v_{1A} + m_B v_{2A} \\ m_A v_{1A} - m_B v_{1A} &= m_A v_{2A} + m_B v_{2A} \\ v_{2A} &= \frac{m_A v_{1A} - m_B v_{1A}}{m_A + m_B} \\ v_{2A} &= \frac{m_A - m_B}{m_A + m_B} v_{1A}\end{aligned}$$

Now, we can plug this value for the final velocity of the moving object back into the velocity relationship we defined earlier.

$$\begin{aligned}v_B &= v_{1A} + \frac{m_A - m_B}{m_A + m_B} v_{1A} \\ v_B &= \frac{m_A + m_B}{m_A + m_B} v_{1A} + \frac{m_A - m_B}{m_A + m_B} v_{1A} \\ v_B &= \frac{2m_A}{m_A + m_B} v_{1A}\end{aligned}$$

 Important

We can interpret this answer in numerous ways.

- Say that the mass of the moving object is much less than the still object. Then, the still object's velocity after the collision is very small.
 - If a ping pong ball collides with a still cannon ball, the cannon ball shouldn't move much.

- If the mass of the moving object is equal to the still object, then the still object's velocity after the collision will be equal to the moving object's initial velocity.
 - In a game of pool, a cue ball of roughly equal mass to the other balls is used to move them.
- If the mass of the moving object is much greater than the still object, the still object's velocity after the collision is very high.
 - If a cannon ball collides with a still ping pong ball, the ping pong ball should move a lot.
- We can heavily simplify calculations for elastic collision problems using the motion of the center of mass, which we will cover in the next two subchapters.

8.4 Center of Mass

- A system's **center of mass** is the average position of all matter in the system.
- In many center of mass problems, we use **point masses**, singular points of some mass m .

Definition 8.4.1 (Center of Mass with Point Masses)

Given point masses m_i of a system, each a distance x_i , y_i , or z_i from the origin, center of masses x_{cm} , y_{cm} , and z_{cm} are defined as follows:

$$x_{\text{cm}} = \frac{\sum_{i=1}^N m_i x_i}{\sum_{i=1}^N m_i} \quad y_{\text{cm}} = \frac{\sum_{i=1}^N m_i y_i}{\sum_{i=1}^N m_i} \quad z_{\text{cm}} = \frac{\sum_{i=1}^N m_i z_i}{\sum_{i=1}^N m_i}$$

Or, condensed into a single vector equation:

$$\vec{R}_{\text{cm}} = \frac{\sum_{i=1}^N m_i \vec{r}}{\sum_{i=1}^N m_i}$$

Definition 8.4.2 (Center of Mass with Continuous Mass Distribution)

Given distance $\vec{r}$ from the origin and total system mass M_{tot} , center of mass $\vec{R}_{\text{cm}}$ is defined as follows:

$$\vec{R}_{\text{cm}} = \frac{1}{M_{\text{tot}}} \int \vec{r} \cdot dm$$

- Because point masses are not given, we integrate with respect to the mass throughout the object.
- To apply this definition, we can find the differential in terms of densities and also integrate over different axes depending on the dimensions involved in the mass distribution.

Linear density, denoted λ :

$$\lambda = \frac{M}{L} = \frac{dm}{dL}$$

$$\therefore dm = \lambda dL = \lambda dx$$

Surface density, denoted σ :

$$\sigma = \frac{M}{A} = \frac{M}{L^2} = \frac{dm}{dA}$$

$$\therefore dm = \sigma dA = \sigma dx dy$$

Volume density, denoted ρ :

$$\rho = \frac{M}{V} = \frac{M}{L^3} = \frac{dm}{dV}$$

$$\therefore dm = \rho dV = \rho dx dy dz$$

Derivation 8.4.3 (Center of Mass of a Uniform Rod)

Intuitively, the center of mass of a uniform rod should be the middle. We can use the center of mass equation for continuous mass distributions to prove this.

We can define the length of any rod as L along the x -axis, and define dm using linear density λ .

$$x_{\text{cm}} = \frac{1}{M_{\text{tot}}} \int x dm$$

$$x_{\text{cm}} = \frac{1}{M_{\text{tot}}} \int x \lambda dL$$

$$x_{\text{cm}} = \frac{\lambda}{M_{\text{tot}}} \int_0^L x dx$$

$$x_{\text{cm}} = \frac{\lambda}{M_{\text{tot}}} \frac{1}{2} x^2 \Big|_0^L$$

$$x_{\text{cm}} = \frac{\lambda}{M_{\text{tot}}} \cdot \frac{L^2}{2}$$

$$x_{\text{cm}} = \frac{M_{\text{tot}}}{L} \cdot \frac{1}{M_{\text{tot}}} \cdot \frac{L^2}{2}$$

$$x_{\text{cm}} = \frac{1}{L} \cdot \frac{L^2}{2}$$

$$x_{\text{cm}} = \frac{L}{2}$$

Derivation 8.4.4 (Center of Mass of a Uniform Cube)

Similar to a uniform rod, the center of mass of a uniform cube is at the very center. Again, we can use the center of mass equation for continuous mass distributions to prove this.

We can define the side lengths of any cube as L and define dm using volume density ρ .

$$\vec{R}_{\text{cm}} = \frac{1}{M_{\text{tot}}} \int \vec{r} \cdot dm$$

Continued on next page

Derivation (Center of Mass of a Uniform Cube *continued*)

$$\begin{aligned}
\vec{R}_{\text{cm}} &= \frac{1}{M_{\text{tot}}} \int \vec{r} \cdot dm \\
\vec{R}_{\text{cm}} &= \frac{1}{M_{\text{tot}}} \iiint \vec{r} \cdot \rho dV \\
\vec{R}_{\text{cm}} &= \frac{\rho}{M_{\text{tot}}} \iiint \vec{r} \cdot dx dy dz \\
\vec{R}_{\text{cm}} &= \frac{\rho}{M_{\text{tot}}} \int_0^L \int_0^L \int_0^L (x\hat{i} + y\hat{j} + z\hat{k}) \cdot dx dy dz \\
\vec{R}_{\text{cm}} &= \frac{\rho}{M_{\text{tot}}} \int_0^L \int_0^L \left(\frac{1}{2}x^2\hat{i} + xy\hat{j} + xz\hat{k} \right) \Big|_{x=0}^{x=L} \cdot dy dz \\
\vec{R}_{\text{cm}} &= \frac{\rho}{M_{\text{tot}}} \int_0^L \int_0^L \left(\frac{L^2}{2}\hat{i} + Ly\hat{j} + Lz\hat{k} \right) \cdot dy dz \\
\vec{R}_{\text{cm}} &= \frac{\rho}{M_{\text{tot}}} \int_0^L \left(\frac{L^2}{2}y\hat{i} + \frac{L}{2}y^2\hat{j} + Lyz\hat{k} \right) \Big|_{y=0}^{y=L} \cdot dz \\
\vec{R}_{\text{cm}} &= \frac{\rho}{M_{\text{tot}}} \int_0^L \left(\frac{L^3}{2}\hat{i} + \frac{L^3}{2}\hat{j} + L^2z\hat{k} \right) \cdot dz \\
\vec{R}_{\text{cm}} &= \frac{\rho}{M_{\text{tot}}} \left(\frac{L^3}{2}z\hat{i} + \frac{L^3}{2}z\hat{j} + \frac{L^2}{2}z^2\hat{k} \right) \Big|_{z=0}^{z=L} \\
\vec{R}_{\text{cm}} &= \frac{\rho}{M_{\text{tot}}} \left(\frac{L^4}{2}\hat{i} + \frac{L^4}{2}\hat{j} + \frac{L^4}{2}\hat{k} \right) \\
\vec{R}_{\text{cm}} &= \frac{M_{\text{tot}}}{V} \cdot \frac{1}{M_{\text{tot}}} \left(\frac{L^4}{2}\hat{i} + \frac{L^4}{2}\hat{j} + \frac{L^4}{2}\hat{k} \right) \\
\vec{R}_{\text{cm}} &= \frac{1}{V} \cdot \left(\frac{L^4}{2}\hat{i} + \frac{L^4}{2}\hat{j} + \frac{L^4}{2}\hat{k} \right) \\
\vec{R}_{\text{cm}} &= \frac{1}{L^3} \cdot \left(\frac{L^4}{2}\hat{i} + \frac{L^4}{2}\hat{j} + \frac{L^4}{2}\hat{k} \right) \\
\vec{R}_{\text{cm}} &= \frac{L}{2}\hat{i} + \frac{L}{2}\hat{j} + \frac{L}{2}\hat{k}
\end{aligned}$$

Motion of the Center of Mass

- Like we've previously established, we can find the velocity and acceleration of the center of mass using derivatives.

Definition 8.4.5 (Velocity of the Center of Mass)

Given point masses m_i of a system each moving with velocity $\vec{v}_i$, the velocity of the center of mass $\vec{v}_{\text{cm}}$ is defined as follows:

$$\vec{v}_{\text{cm}} = \frac{\sum_{i=1}^N m_i \vec{v}_i}{\sum_{i=1}^N m_i}$$

- Now, say the total mass of the system is $\sum_{i=1}^N m_i = M_{\text{tot}}$. If we multiply the velocity of the center of mass by the total mass, we arrive at a new definition for total momentum $\vec{P}$:

$$M_{\text{tot}} \vec{v}_{\text{cm}} = \sum_{i=1}^N m_i \vec{v}_i = m_1 \vec{v}_1 + m_2 \vec{v}_2 + \cdots + m_N \vec{v}_N = \vec{P}$$

- Thus, a system with a net external force of zero has a center of mass that moves with constant velocity.
- Furthermore, if the center of mass is already at rest, then it stays at rest when $\Sigma \vec{F} = 0$.

Definition 8.4.6 (Acceleration of the Center of Mass)

Given point masses m_i of a system each with acceleration $\vec{a}_i$, the acceleration of the center of mass $\vec{a}_{\text{cm}}$ is defined as follows:

$$\vec{a}_{\text{cm}} = \frac{\sum_{i=1}^N m_i \vec{a}_i}{\sum_{i=1}^N m_i}$$

- Now, it follows that multiplying the acceleration of the center of mass by the system's total mass yields the rate of change of the system's total momentum: the *net force*.

$$M_{\text{tot}} \vec{a}_{\text{cm}} = \sum_{i=1}^N m_i \vec{a}_i = m_1 \vec{a}_1 + m_2 \vec{a}_2 + \cdots + m_N \vec{a}_N = \Sigma \vec{F}$$

- However, due to Newton's third law, the net internal forces cancel out, leaving just the net external forces. Thus:

$$M_{\text{tot}} \vec{a}_{\text{cm}} = \Sigma \vec{F} = \Sigma \vec{F}_{\text{ext}} + \Sigma \vec{F}_{\text{int}} = \Sigma \vec{F}_{\text{ext}}$$

- Now, imagine a scenario where a shell is shot into the air, following a parabolic trajectory. Even when it splits into multiple fragments that each follow their own trajectory, the center of mass *follows the original trajectory* as if the masses of the system were all concentrated on that path.
- This explains the definition of net force as shown in Chapter 4.

Example 8.4.7 (Romeo and Juliet)

Romeo, who is sitting in the rear of their boat in still water, entertains Juliet by playing his guitar. After the serenade, Juliet, who is sitting at the front of the boat, moves to the rear to kiss Romeo. The 80-kg boat is facing the shore and the 55-kg Juliet needs to move 2.7 m to reach the 77-kg Romeo. How far does the boat move and in what direction?

Now, the center of mass of the system does not change because the net external force acting on the system is zero. Now, it follows that the boat is $L = 2.7$ m long, and that its center of mass is at half that length. Now, the final positions relative to the origin all move by the distance d toward the shore to compensate for Juliet moving toward the boat's rear, which is away from the shore. We will choose the front of the boat as the origin.

$$\begin{aligned}
 x_{\text{cm}-i} &= x_{\text{cm}-f} \\
 \frac{m_J x_{Ji} + m_R x_{Ri} + m_B x_{Bi}}{m_J + m_R + m_B} &= \frac{m_J x_{Jf} + m_R x_{Rf} + m_B x_{Bf}}{m_J + m_R + m_B} \\
 m_J x_{Ji} + m_R x_{Ri} + m_B x_{Bi} &= m_J x_{Jf} + m_R x_{Rf} + m_B x_{Bf} \\
 m_R L + m_B \frac{L}{2} &= m_J (L - d) + m_R (L - d) + m_B \left(\frac{L}{2} - d \right) \\
 m_R L + m_B \frac{L}{2} &= \left(m_J + m_R + \frac{m_B}{2} \right) L - (m_J + m_R + m_B) d \\
 (m_J + m_R + m_B) d &= \left(m_J + m_R + \frac{m_B}{2} \right) L - m_R L - m_B \frac{L}{2} \\
 d &= \frac{\left(m_J + m_R + \frac{m_B}{2} \right) L - m_R L - m_B \frac{L}{2}}{m_J + m_R + m_B} \\
 d &= \frac{m_J L}{m_J + m_R + m_B} \\
 d &= \frac{(55 \text{ kg})(2.7 \text{ m})}{55 \text{ kg} + 77 \text{ kg} + 80 \text{ kg}} \\
 d &= 0.7 \text{ m}
 \end{aligned}$$

The boat moves 0.7 m toward the shore.

8.5 Center-of-Mass Frame

- By using the center of mass as a reference frame, elastic collision problems can be heavily simplified.
 - Known as the **Center-of-Mass reference frame**, or just **CM reference frame**.
- In the CM reference frame, *the velocity of the center of mass is zero and the total momentum is zero.*

Definition 8.5.1 (Center-of-Mass Frame)

Given that superscript * denotes “in the reference frame of the Center of Mass” and given any object A:

$$v_i^* = -v_f^*$$

$$v_A^* = v_{A,\text{lab}} + v_{\text{lab},\text{cm}}$$

$$v_A^* = v_{A,\text{lab}} - v_{\text{cm}}$$



- Another use case of the CM reference frame is for defining the kinetic energy of a system.

Definition 8.5.2 (Kinetic Energy of a System)

Given masses of the system m_i each with velocity v_i^* :

$$K_{\text{sys}} = \sum_{i=1}^N \frac{1}{2} m_i v_i^{*2} + \frac{1}{2} M_{\text{tot}} v_{\text{cm}}^2$$

$$K_{\text{sys}} = K_{\text{rel}} + K_{\text{cm}}$$

**Example 8.5.3**

Initially, block 1 with mass $m_1 = 1$ kg is moving horizontally to the right with velocity $v_1 = 1$ m/s and block 2 with mass $m_2 = 0.5$ kg is at rest. Block 1 collides elastically with block 2, and after that, block 2 collides elastically with a wall to the right. What impulse does the wall give to block 2?

Because there are no external forces in the system before block 2 collides with the wall, we can calculate the velocity of the center of mass at the time when block 2 is at rest, as it will remain constant until the collision between block 2 and the wall.

$$v_{\text{cm}} = \frac{m_1 v_1}{m_1 + m_2}$$

$$v_{\text{cm}} = \frac{(1 \text{ kg})(1 \text{ m/s})}{1 \text{ kg} + 0.5 \text{ kg}}$$

$$v_{\text{cm}} = \frac{2}{3} \text{ m/s}$$

Now that we know the velocity of the center of mass, we can use the CM reference frame to find the velocity of block 2 after the first collision. First, we find the initial velocity of block 2 in the CM reference frame.

Continued on next page



Example (Elastic Blocks *continued*)

$$v_2^* = v_2 + v_{\text{lab,cm}}$$

$$v_2^* = v_2 - v_{\text{cm}}$$

$$v_2^* = 0 \text{ m/s} - \frac{2}{3} \text{ m/s}$$

$$v_2^* = -\frac{2}{3} \text{ m/s}$$

Now, we use the fact that an object reverses its direction but maintains the same speed relative to the center of mass after an elastic collision.

$$v_{2f}^* = -v_2^*$$

$$v_{2f}^* = -\left(-\frac{2}{3} \text{ m/s}\right)$$

$$v_{2f}^* = \frac{2}{3} \text{ m/s}$$

To find the velocity of block 2 after the first collision, we can apply relative velocities again.

$$v_{2f} = v_{2f}^* + v_{\text{cm}}$$

$$v_{2f} = \frac{2}{3} \text{ m/s} + \frac{2}{3} \text{ m/s}$$

$$v_{2f} = \frac{4}{3} \text{ m/s}$$

Now, it follows that the second collision is also elastic. Intuitively, block 2's velocity after the second collision has the same magnitude, albeit in the opposite direction. Now that we know the velocity of block 2 just before and just after its collision with the wall, we can calculate the impulse exerted by the wall on block 2.

$$J = \Delta p$$

$$J = m_2(v_{2ff} - v_{2f})$$

$$J = (0.5 \text{ kg})\left(-\frac{4}{3} \text{ m/s} - \frac{4}{3} \text{ m/s}\right)$$

$$J = -\frac{4}{3} \text{ kg} \cdot \text{m/s}$$



9 Rotation of Rigid Bodies

- **Rigid bodies** are objects that do not experience deformations from forces.
- They are idealized models that will be used throughout this chapter to simplify calculations for rotational motion.

9.1 Angular Kinematics

- Imagine a rigid body rotating about a fixed axis. We can describe its motion by picking some point P on the body and using the angle θ it makes with $+x$ -axis to describe rotational position.
- **Angular displacement** is measured in *radians* (rad) rather than degrees.
 - Denoted θ .

$$1 \text{ rad} \cdot \frac{360^\circ}{2\pi \text{ rad}} \approx 57.296^\circ$$

- We define one radian as the angle of an arc s given that $s = \text{radius } r$.

$$s = \theta r \text{ or } \theta = \frac{s}{r}$$

- **Angular velocity** is the rate of change of angular displacement, measured in radians per second.
 - Denoted ω .
 - Note that while points farther from the axis of rotation have a larger displacement over the same time interval, *all points on a rigid body have the same angular velocity*.
- **Angular acceleration** is the rate of change of angular velocity, measured in radians per second squared.
 - Denoted α .
- The standard for rotational motion is that counterclockwise from the $+x$ -axis is the positive direction while clockwise from the $+x$ -axis is the negative direction.
- Applying our definition of angular displacement, we can find the magnitude of the tangential speed and acceleration:

$$\left| \frac{ds}{dt} \right| = \left| \frac{d\theta}{dt} \right| r$$
$$v_t = \omega r$$

$$\frac{dv_t}{dt} = \frac{d\omega}{dt} r$$
$$a_t = \alpha r$$

- Using this new definition for linear speed, we can express centripetal acceleration in terms of angular speed:

$$a_c = \omega^2 r$$

- Note that because ω is angular speed, α is the rate of change of the angular speed. We can denote the angular velocity and acceleration around some rotational axis z :

$$\begin{aligned}\Delta\theta &= \theta_2 - \theta_1 \\ \omega_{\text{av-}z} &= \frac{\Delta\theta}{\Delta t} = \frac{2\pi}{T} = 2\pi f \\ \omega_z &= \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} \\ \alpha_{\text{av-}z} &= \frac{\Delta\omega_z}{\Delta t} \\ \alpha_z &= \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega_z}{\Delta t} = \frac{d\omega_z}{dt} = \frac{d^2\theta}{dt^2}\end{aligned}$$

Definition 9.1.1 (Angular Kinematic Equations)

Given axis of rotation z and constant angular acceleration, the following equations hold:

$$\begin{aligned}\omega_z &= \omega_{0z} + \alpha_z \Delta t \\ \Delta\theta &= \omega_{0z} \Delta t + \frac{1}{2} \alpha_z \Delta t^2 \\ \omega_z^2 &= \omega_{0z}^2 + 2\alpha_z \Delta\theta\end{aligned}$$

- These are analogous to the kinematic equations we learned in Chapter 2.
- We defined these angular kinematic equations along the z -axis because $\|\vec{\theta}\|$, $\|\vec{\omega}\|$, and $\|\vec{\alpha}\|$ are all equal to their z -components since the axis of rotation is fixed.
- Consequently, $\vec{\theta}$, $\vec{\omega}$, and $\vec{\alpha}$ are all parallel to the axis of rotation, or in other words, *perpendicular to the plane of rotation*. Specifically, their direction is denoted by the right-hand rule.

Important

The **right-hand** rule is used to denote the direction of vector products. In this case, you curl your fingers in the direction of rotation (clockwise or counterclockwise) and stick your thumb out. *The direction of your thumb is analogous to the direction of $\vec{\omega}$ and $\vec{\alpha}$.*

Example 9.1.2 (Throwing a Discus)

An athlete spins a discus in a circle of radius $r = 80.0$ cm. At a particular instant, the athlete rotates at $\omega = 10.0$ rad/s, and that angular speed changes at a rate of 50.0 rad/s². At this instant, find the tangential and centripetal components of the discus' acceleration, then use those components to find the magnitude.

Because we are given the angular speed and acceleration, we are able to find the components of the discus' acceleration using the formulas we just established.

Continued on next page

Example (Throwing a Discus *continued*)

$$a_c = \omega^2 r$$

$$a_c = (10 \text{ rad/s})^2 \left(80 \text{ cm} \cdot \frac{1 \text{ m}}{100 \text{ cm}} \right)$$

$$a_c = 80.0 \text{ m/s}^2$$

$$a_t = \alpha r$$

$$a_t = (50.0 \text{ rad/s}^2) \left(80 \text{ cm} \cdot \frac{1 \text{ m}}{100 \text{ cm}} \right)$$

$$a_t = 40.0 \text{ m/s}^2$$

Now, it follows that these are the components of the discus' acceleration.

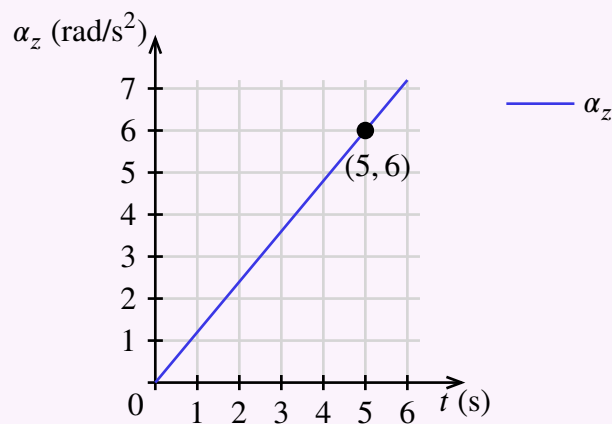
$$a = \sqrt{a_c^2 + a_t^2}$$

$$a = \sqrt{(80 \text{ m/s}^2)^2 + (40 \text{ m/s}^2)^2}$$

$$a = 89.4 \text{ m/s}^2$$

Example 9.1.3 (Variable-Speed Motor)

A technician is testing a computer-controlled, variable-speed motor. She attaches a thin disk to the motor shaft, with the shaft at the center of the disk. The disk starts from rest, and sensors attached to the motor shaft measure its angular acceleration α_z as a function of time.



The results from one test run of the motor.

- 1) Through how many revolutions has the disk turned in the first 5.0 s?
- 2) What is the angular velocity of the disk at $t = 5.0$ s?
- 3) What is the angular velocity of the disk when it has turned through 2.00 rev?

1) Because we know the disk starts from rest, we can continuously find antiderivatives.

$$\alpha_z(t) = \left(\frac{6 \text{ rad/s}^2}{5 \text{ s}} \right) t$$

Continued on next page

Example 9.1.4 (Variable-Speed Motor *continued*)

$$\alpha_z(t) = (1.2 \text{ rad/s}^3)t$$

$$\omega_z(t) = \int (1.2 \text{ rad/s}^3)t \, dt$$

$$\omega_z(t) = (0.6 \text{ rad/s}^3)t^2$$

$$\theta(t) = \int (0.6 \text{ rad/s}^3)t^2 \, dt$$

$$\theta(t) = (0.2 \text{ rad/s}^3)t^3$$

$$\theta(5 \text{ s}) = (0.2 \text{ rad/s}^3)(5 \text{ s})^3$$

$$\theta(5 \text{ s}) = 25 \text{ rad}$$

$$N_{\text{revs}} = 25 \text{ rad} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}}$$

$$N_{\text{revs}} = \mathbf{4.0 \text{ rev}}$$

2) We could integrate $\alpha_z(t)$ from $t = 0$ to $t = 5$, but we've already found $\omega_z(t)$ during our steps to find $\theta(t)$.

$$\omega_z(t) = (0.6 \text{ rad/s}^3)t^2$$

$$\omega_z(5 \text{ s}) = (0.6 \text{ rad/s}^3)(5 \text{ s})^2$$

$$\omega_z(5 \text{ s}) = \mathbf{15 \text{ rad/s}}$$

3) We first need to convert the revolutions to radians.

$$\theta = 2.00 \text{ rev} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} = 4\pi \text{ rad}$$

Now, we can solve for the time when 2 revolutions have passed using $\theta(t)$, which we found in part 1.

$$\theta(t) = (0.2 \text{ rad/s}^3)t^3$$

$$4\pi \text{ rad} = (0.2 \text{ rad/s}^3)t^3$$

$$t = \sqrt[3]{\frac{4\pi \text{ rad}}{0.2 \text{ rad/s}^3}}$$

$$t = 3.98 \text{ s}$$

This value can then be plugged into $\omega_z(t)$.

$$\omega_z(t) = (0.6 \text{ rad/s}^3)t^2$$

$$\omega_z(3.98 \text{ s}) = (0.6 \text{ rad/s}^3)(3.98 \text{ s})^2$$

$$\omega_z(3.98 \text{ s}) = \mathbf{9.5 \text{ rad/s}}$$



Example 9.1.5 (Electric Motor Rotation)

At $t = 0$, the current to a DC electric motor is reversed, resulting in an angular displacement of the motor shaft given by $\theta(t) = (260 \text{ rad/s})t - (19.2 \text{ rad/s}^2)t^2 - (1.56 \text{ rad/s}^3)t^3$

- 1) At what time is the angular velocity of the motor shaft zero?
- 2) Calculate the angular acceleration at the instant that the motor shaft has zero angular velocity.
- 3) How many revolutions does the motor shaft turn through between the time the current is reversed and the instant the angular velocity is zero?
- 4) How fast was the motor shaft rotating at $t = 0$ when the current was reversed?
- 5) Calculate the average angular velocity from $t = 0$ to when the motor shaft's angular velocity is zero.

1) We can simply find $\theta'(t)$ and find a valid zero of the resulting function.

$$\theta'(t) = \frac{d}{dt}((260 \text{ rad/s})t - (19.2 \text{ rad/s}^2)t^2 - (1.56 \text{ rad/s}^3)t^3)$$

$$\omega(t) = 260 \text{ rad/s} - 2(19.2 \text{ rad/s}^2)t - 3(1.56 \text{ rad/s}^3)t^2$$

$$0 = 260 \text{ rad/s} - 2(19.2 \text{ rad/s}^2)t - 3(1.56 \text{ rad/s}^3)t^2$$

$$t = \frac{38.4 \pm \sqrt{(-38.4)^2 - 4(-4.68)(260)}}{2(-4.68)}$$

$$t = -11.6 \text{ s}$$

$$t = 4.41 \text{ s}$$

2) Similarly, we can just find $\omega'(t)$, then use the time we just calculated.

$$\omega'(t) = \frac{d}{dt}(260 \text{ rad/s} - 2(19.2 \text{ rad/s}^2)t - 3(1.56 \text{ rad/s}^3)t^2)$$

$$\alpha(t) = 38.4 \text{ rad/s}^2 - 6(1.56 \text{ rad/s}^3)t$$

$$\alpha(4.41 \text{ s}) = 38.4 \text{ rad/s}^2 - 6(1.56 \text{ rad/s}^3)(4.41 \text{ s})$$

$$\alpha(4.41 \text{ s}) = -79.6 \text{ rad/s}^2$$

3) Again, we already calculated this time interval. Thus, we just need to use $\theta(t)$.

$$\theta(4.41 \text{ s}) = (260 \text{ rad/s})(4.41 \text{ s}) - (19.2 \text{ rad/s}^2)(4.41 \text{ s})^2 - (1.56 \text{ rad/s}^3)(4.41 \text{ s})^3$$

$$\theta(4.41 \text{ s}) = 639 \text{ rad}$$

$$\theta(4.41 \text{ s}) = 639 \text{ rad} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}}$$

$$\theta(4.41 \text{ s}) = 102 \text{ rev}$$

Continued on next page



Example (Electric Motor Rotation *continued*)

4) Since we've already found $\omega(t)$, we can just find its value at $t = 0$.

$$\omega(0) = 260 \text{ rad/s}$$

5) Because we are calculating the average angular velocity, we use $\theta(t)$.

$$\omega_{\text{av}} = \frac{\theta(4.41 \text{ s}) - \theta(0 \text{ s})}{4.41 \text{ s} - 0 \text{ s}}$$

$$\omega_{\text{av}} = 145 \text{ rad/s}$$

9.2 Energy in Rotational Motion

- **Moment of inertia** is an object's resistance to angular acceleration, measured in ($\text{kg} \cdot \text{m}^2$).
 - Denoted I .

Definition 9.2.1 (Moment of Inertia with Point Masses)

Given point masses m_i on a rigid body, each a distance r_i from the axis of rotation, let moment of inertia be defined as follows:

$$I = \sum_{i=1}^N m_i r_i^2$$

- Moment of inertia is applied in a definition for **rotational kinetic energy** K of a rigid body.

Definition 9.2.2 (Rotational Kinetic Energy of a Rigid Body)

Given moment of inertia for some rotational axis I and angular speed of rigid body ω , let rotational kinetic energy be defined as follows:

$$K_{\text{rot}} = \frac{1}{2} I \omega^2$$

Note

This is still kinetic energy, and is the sum of the kinetic energies of all points within a rotating rigid bodies. There will be problems where an object has both translational and rotational kinetic energy.

- According to this equation, the more resistant an object is to rotational acceleration, the greater its kinetic energy for some given ω .

Definition 9.2.3 (Moment of Inertia with Continuous Mass Distributions)

Given the distance of each infinitesimally small mass on the rigid body from the axis of rotation r , let moment of inertia be defined as follows:

$$I = \int r^2 dm$$



- Thus, the axis of rotation chosen can change the calculated moment of inertia.
- Using this formula, we can derive the moments of inertia for common forms.
 - **Slender rod with an axis of rotation through the center ($\perp$ to the rod's surface):**

$$I_{\text{rod-cm}} = \frac{1}{12} M_{\text{tot}} L^2$$

- **Slender rod with an axis of rotation at one end ($\perp$ to the rod's surface):**

$$I_{\text{rod-end}} = \frac{1}{3} M_{\text{tot}} L^2$$

- **Rectangular plate with an axis of rotation through the center:**

$$I_{\text{rect-cm}} = \frac{1}{12} M_{\text{tot}} (a^2 + b^2)$$

- **Thin rectangular plate with an axis of rotation along an edge:**

$$I_{\text{rect-edge}} = \frac{1}{3} M_{\text{tot}} a^2$$

- **Hollow/solid cylinder:**

$$I_{\text{cylinder}} = \frac{1}{2} M_{\text{tot}} (R_1^2 + R_2^2)$$

- **Thin-walled hollow cylinder:**

$$I_{\text{thin-cylinder}} = M_{\text{tot}} R^2$$

- **Solid sphere:**

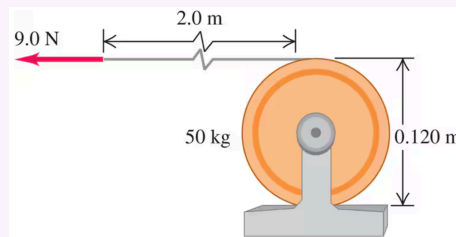
$$I_{\text{sphere}} = \frac{2}{5} M_{\text{tot}} R^2$$

- **Thin-walled hollow sphere:**

$$I_{\text{solid-sphere}} = \frac{2}{3} M_{\text{tot}} R^2$$

Example 9.2.4 (Unwinding Cable)

A light, nonstretching cable is wrapped around a solid cylinder of mass 50 kg and diameter 0.120 m that rotates in frictionless bearings about a stationary horizontal axis. The cable's free end is pulled with a constant 9.0 N force for a distance of 2.0 m, turning the cylinder as it unwinds without slipping. The cylinder is initially at rest. Find its final angular speed and the final speed of the cable.



Because we are just given the total mass of the solid cylinder, we can use the derived formula for a cylinder's moment of inertia.

$$I = \frac{1}{2}mR^2$$

$$I = \frac{1}{2}(50 \text{ kg})(0.12 \text{ m} \cdot 0.5)^2$$

$$I = 0.090 \text{ kg} \cdot \text{m}^2$$

Now that we have the cylinder's moment of inertia, we can solve for the cylinder's final angular speed using the work-energy theorem and rotational kinetic energy.

$$W = \Delta K$$

$$Fd \cos 90^\circ = \frac{1}{2}I\omega^2$$

$$\frac{2Fd}{I} = \omega^2$$

$$\omega = \sqrt{\frac{2Fd}{I}}$$

$$\omega = \sqrt{\frac{2 \cdot 9 \text{ N} \cdot 2 \text{ m}}{0.090 \text{ kg} \cdot \text{m}^2}}$$

$$\omega = 20 \text{ rad/s}$$

It follows that we can use the cylinder's final angular speed to find the cylinder's final linear speed.

$$v = \omega R$$

$$v = (20 \text{ rad/s})(0.12 \text{ m} \cdot 0.5)$$

$$v = 1.2 \text{ m/s}$$

Example 9.2.5 (Rotating Pulley)

Consider a system with two blocks. Block A sits on a frictionless tabletop and is attached to a light rope that goes over a pulley at the edge of the table. Block B hangs from the other end of the rope with a mass of $m_B = 5.00$ kg. The pulley has a radius of $r = 0.200$ m and a moment of inertia of $I = 1.30$ kg $\cdot$ m². If the pulley is rotating with an angular speed of $\omega = 8.00$ rad/s after block B has descended 1.20 m, what is block A's mass?

Because we are given the pulley's angular speed and radius, we can find its linear speed after block B has descended by 1.2 m.

$$\begin{aligned}v &= \omega r \\v &= (8 \text{ rad/s})(0.2 \text{ m}) \\v &= 1.6 \text{ m/s}\end{aligned}$$

It follows that the linear speed of the pulley is equal to the speeds of blocks A and B at any given time. Thus, we can use conservation of energy to find the mass of block A.

$$\begin{aligned}E_1 &= E_2 \\U_A + U_B &= U_A + K_{\text{pulley}} + K_A + K_B \\m_B gh &= \frac{1}{2} I \omega^2 + \frac{1}{2} m_A v^2 + \frac{1}{2} m_B v^2 \\-\frac{1}{2} m_A v^2 &= \frac{1}{2} I \omega^2 + \frac{1}{2} m_B v^2 - m_B gh \\m_A &= \frac{2m_B gh - I \omega^2 - m_B v^2}{v^2} \\m_A &= \frac{2(5 \text{ kg})(9.81 \text{ m/s})(1.2 \text{ m}) - (1.3 \text{ kg} \cdot \text{m}^2)(8 \text{ rad/s})^2 - (5 \text{ kg})(1.6 \text{ m/s})^2}{(1.6 \text{ m/s})^2} \\m_A &= 8.48 \text{ kg}\end{aligned}$$

Gravitational Potential Energy of Extended Bodies

Definition 9.2.6 (Gravitational Potential Energy)

Given total mass of extended body M_{tot} and vertical coordinate y_{cm} , let gravitational potential energy be defined as follows:

$$U = M_{\text{tot}} g y_{\text{cm}}$$

- This formula is the same as finding the gravitational potential energy of the masses on an extended body and summing them up.

9.3 Parallel Axis Theorem

Theorem 9.3.1 (Parallel Axis Theorem)

Let I_P denote the moment of inertia about a rotational axis through point P , and let I_{cm} denote the moment of inertia about a parallel rotational axis through the center of mass. Additionally, let M_{tot} be the rigid body's total mass and d be the distance between the parallel axes.

$$I_P = I_{\text{cm}} + M_{\text{tot}}d^2$$

- The main application for this formula is for finding moments of inertia of known shapes on a rotational axis *offset* some distance d from the rotational axis through the center of mass.

9.4 Moment-of-Inertia Derivations

This subchapter will include the derivations for the moments of inertia of different forms given in subchapter 9.2.

Derivation 9.4.1 (Rod at one end)

We can start by establishing a relationship between a rod's linear density and the differential dm . Let L be the length of the rod.

$$\lambda = \frac{M_{\text{tot}}}{L} = \frac{dm}{dL}$$

$$\therefore dm = \lambda dL$$

Now, we can apply the moment of inertia formula.

$$I_{\text{rod-end}} = \int r^2 dm$$

$$I_{\text{rod-end}} = \int r^2 \lambda dL$$

$$I_{\text{rod-end}} = \lambda \int_0^L r^2 dr$$

$$I_{\text{rod-end}} = \lambda \left(\frac{1}{3} r^3 \right) \Big|_0^L$$

$$I_{\text{rod-end}} = \lambda \left(\frac{1}{3} L^3 \right)$$

$$I_{\text{rod-end}} = \left(\frac{M_{\text{tot}}}{L} \right) \left(\frac{1}{3} L^3 \right)$$

$$I_{\text{rod-end}} = \frac{1}{3} M_{\text{tot}} L^2$$

Derivation 9.4.2 (Rod at the center)

We can start by establishing a relationship between a rod's linear density and the differential dm . Let L be the length of the rod.

$$\lambda = \frac{M_{\text{tot}}}{L} = \frac{dm}{dL}$$

$$\therefore dm = \lambda dL$$

Now, we can apply the moment of inertia formula.

$$I_{\text{rod-cm}} = \int r^2 dm$$

$$I_{\text{rod-cm}} = \int r^2 \lambda dL$$

$$I_{\text{rod-cm}} = \lambda \int_{-L/2}^{L/2} r^2 dr$$

$$I_{\text{rod-cm}} = \lambda \left(\frac{1}{3} r^3 \right) \Big|_{-L/2}^{L/2}$$

$$I_{\text{rod-cm}} = \lambda \left(\frac{1}{3} \left(\frac{L}{2} \right)^3 - \frac{1}{3} \left(-\frac{L}{2} \right)^3 \right)$$

$$I_{\text{rod-cm}} = \lambda \left(\frac{1}{12} L^3 \right)$$

$$I_{\text{rod-cm}} = \left(\frac{M_{\text{tot}}}{L} \right) \left(\frac{1}{12} L^3 \right)$$

$$I_{\text{rod-cm}} = \frac{1}{12} M_{\text{tot}} L^2$$

Another option is using the parallel axis theorem, given that we just derived $I_{\text{rod-end}}$.

$$I_{\text{rod-end}} = I_{\text{rod-cm}} + M_{\text{tot}} d^2$$

$$I_{\text{rod-cm}} = I_{\text{rod-end}} - M_{\text{tot}} d^2$$

$$I_{\text{rod-cm}} = \frac{1}{3} M_{\text{tot}} L^2 - M_{\text{tot}} \left(\frac{L}{2} \right)^2$$

$$I_{\text{rod-cm}} = \frac{4}{12} M_{\text{tot}} L^2 - \frac{3}{12} M_{\text{tot}} L^2$$

$$I_{\text{rod-cm}} = \frac{1}{12} M_{\text{tot}} L^2$$

Derivation 9.4.3 (Cylinder)

We can start by establishing a relationship between a cylinder's volume density and the differential dm . Let r_2 be the outer radius and r_1 be the inner radius.

$$\rho = \frac{M_{\text{tot}}}{V} = \frac{M_{\text{tot}}}{\pi(r_2^2 - r_1^2)L} = \frac{dm}{dV} = \frac{dm}{2\pi r L dr}$$

$$\therefore dm = \rho dV = \rho(2\pi r L dr)$$

Now, we can apply the moment of inertia formula.

$$I_{\text{cylinder}} = \int r^2 dm$$

$$I_{\text{cylinder}} = \int r^2 \rho dV$$

$$I_{\text{cylinder}} = \int r^2 \rho(2\pi r L dr)$$

$$I_{\text{cylinder}} = \rho 2\pi L \int_{r_1}^{r_2} r^3 dr$$

$$I_{\text{cylinder}} = \rho 2\pi L \left(\frac{1}{4} r^4 \right) \Big|_{r_1}^{r_2}$$

$$I_{\text{cylinder}} = \frac{1}{2} \rho \pi L (r_2^4 - r_1^4)$$

$$I_{\text{cylinder}} = \frac{1}{2} \left(\frac{M_{\text{tot}}}{\pi(r_2^2 - r_1^2)L} \right) \pi L (r_2^2 + r_1^2)(r_2^2 - r_1^2)$$

$$I_{\text{cylinder}} = \frac{1}{2} M_{\text{tot}} (r_2^2 + r_1^2)$$



Derivation 9.4.4 (Thin-walled Cylinder)

Because the cylinder is infinitely thin, we will relate dm to surface density σ . Let R be the radius of the cylinder.

$$\sigma = \frac{M_{\text{tot}}}{A} = \frac{M_{\text{tot}}}{2\pi RL} = \frac{dm}{dA}$$

$$\therefore dm = \sigma dA = \sigma 2\pi R dx$$

Now, we can apply the moment of inertia formula.

$$I_{\text{thin-cylinder}} = \int R^2 dm$$

$$I_{\text{thin-cylinder}} = \int R^2 \sigma dA$$

$$I_{\text{thin-cylinder}} = \int R^2 \sigma (2\pi R dx)$$

$$I_{\text{thin-cylinder}} = \sigma 2\pi R^3 \int_0^L dx$$

$$I_{\text{thin-cylinder}} = \sigma 2\pi R^3 x \Big|_0^L$$

$$I_{\text{thin-cylinder}} = \sigma 2\pi R^3 (L)$$

$$I_{\text{thin-cylinder}} = \left(\frac{M_{\text{tot}}}{2\pi RL} \right) 2\pi R^3 (L)$$

$$I_{\text{thin-cylinder}} = M_{\text{tot}} R^2$$

Derivation 9.4.5 (Solid Sphere)

Imagine that a solid sphere of radius R is divided into infinitesimally thin disks of radius r . Now, let x be the distance of each disk's center from the sphere's center. For each disk, we can form a right triangle between the sphere's center, the disk's center, and a point on the disk's edge. Thus, we can express each disk's radius r as follows:

$$\begin{aligned}x^2 + r^2 &= R^2 \\r^2 &= R^2 - x^2\end{aligned}$$

When establishing a relationship between a solid sphere's volume density and the differential dm , we can use the volumes of the disks, which are essentially cylinders, each with a length of dx .

$$\begin{aligned}\rho &= \frac{M_{\text{tot}}}{V} = \frac{M_{\text{tot}}}{\frac{4}{3}\pi R^3} = \frac{dm}{dV} = \frac{dm}{\pi r^2 dx} = \frac{dm}{\pi(R^2 - x^2) dx} \\ \therefore dm &= \rho dV = \rho \pi (R^2 - x^2) dx\end{aligned}$$

Now, we can apply the moment of inertia formula.

$$\begin{aligned}I_{\text{solid-sphere}} &= \int r^2 dm \\ I_{\text{solid-sphere}} &= \int (R^2 - x^2) dm \\ I_{\text{solid-sphere}} &= \int (R^2 - x^2) \rho dV \\ I_{\text{solid-sphere}} &= \int_0^R (R^2 - x^2) \rho \pi (R^2 - x^2) dx \\ I_{\text{solid-sphere}} &= \rho \pi \int_0^R (R^4 - 2R^2x^2 + x^4) dx \\ I_{\text{solid-sphere}} &= \rho \pi \left(R^4x - \frac{2}{3}R^2x^3 + \frac{1}{5}x^5 \right) \Big|_0^R \\ I_{\text{solid-sphere}} &= \rho \pi \left(\frac{8}{15}R^5 \right) \\ I_{\text{solid-sphere}} &= \left(\frac{M_{\text{tot}}}{\frac{4}{3}\pi R^3} \right) \pi \left(\frac{8}{15}R^5 \right) \\ I_{\text{solid-sphere}} &= \frac{2}{5}M_{\text{tot}}R^2\end{aligned}$$

Derivation 9.4.6 (Thin-walled Sphere)

Because the sphere is infinitely thin, we will relate dm to surface density σ . We can use spherical coordinates where the radial distance is a constant $\rho = R$, the polar angle is θ , and the azimuthal angle is ϕ .

$$\sigma = \frac{M_{\text{tot}}}{A} = \frac{M_{\text{tot}}}{4\pi R^2} = \frac{dm}{dA}$$

$$\therefore dm = \sigma dA = \sigma R^2 \sin(\phi) d\theta d\phi$$

Now, we can find the distance from the axis in terms of the sphere's radius R using the azimuthal angle.

$$\cos\left(\frac{\pi}{2} - \phi\right) = \sin(\phi) = \frac{r}{R}$$

$$\therefore r = R \sin(\phi)$$

Now, we can apply the moment of inertia formula.

$$I_{\text{thin-sphere}} = \int r^2 dm$$

$$I_{\text{thin-sphere}} = \int R^2 \sin^2(\phi) dm$$

$$I_{\text{thin-sphere}} = \iint R^2 \sin^2(\phi) (\sigma R^2 \sin(\phi) d\theta d\phi)$$

$$I_{\text{thin-sphere}} = \sigma \int_0^{2\pi} d\theta \int_0^\pi R^4 \sin^3(\phi) d\phi$$

$$I_{\text{thin-sphere}} = \sigma \int_0^{2\pi} d\theta \cdot \left(\frac{R^4}{3} \cos^3(\phi) - R^4 \cos(\phi) \right) \Big|_{\phi=0}^{\phi=\pi}$$

$$I_{\text{thin-sphere}} = \sigma \left(\frac{4}{3} R^4 \right) \int_0^{2\pi} d\theta$$

$$I_{\text{thin-sphere}} = \sigma \left(\frac{4}{3} R^4 \right) (2\pi)$$

$$I_{\text{thin-sphere}} = \left(\frac{M_{\text{tot}}}{4\pi R^2} \right) \left(\frac{4}{3} R^4 \right) (2\pi)$$

$$I_{\text{thin-sphere}} = \frac{2}{3} M_{\text{tot}} R^2$$

10 Dynamics of Rotational Motion

- We established in earlier chapters that forces induce acceleration in objects.
- It follows that forces also induce angular acceleration in objects, but they specifically have to cause twisting or turning.

10.1 Torque

- **Torque** is the tendency of a force to cause or change's an object's rotational motion, measured in newton-meters ($\text{N} \cdot \text{m}$).
 - Denoted τ .
 - Torque is a vector quantity in the same direction as the $\vec{\alpha}$.
 - While this is technically the same unit used by energy, the context is different.
- Torque is caused by a force acting along its **line of action** on an object some perpendicular distance from the reference point known as the **lever arm** or **moment arm**.

Definition 10.1.1 (Torque)

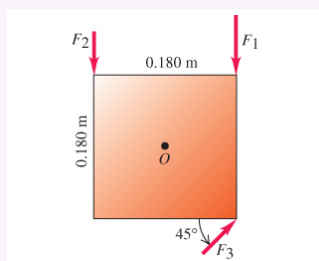
Given force $\vec{F}$ and lever arm $\vec{r}$, let torque be defined as follows:

$$\vec{\tau} = \vec{r} \times \vec{F} = rF \sin(\theta) \hat{n}_{\perp}$$
$$\tau = rF \sin(\theta)$$

- Generally, torque that causes counterclockwise rotation is positive while torque that causes clockwise rotation is negative.
- Similar to other angular quantities, we often refer to torque as its component along the rotational axis, usually the z-axis.

Example 10.1.2 (Rotating a Square Plate)

A square metal plate with 0.180 m side lengths is pivoted about an axis through point O at its center and perpendicular to the plate. Calculate the net torque about this axis due to the three forces shown in the figure given $F_1 = 28.0 \text{ N}$, $F_2 = 14.9 \text{ N}$, and $F_3 = 16.1 \text{ N}$.



Continued on next page

Example (Rotating a Square Plate *continued*)

Because the lever arm is a perpendicular distance from the rotational axis, we need to calculate the diagonal distance from point O to the square's corners.

$$(0.09 \text{ m})^2 + (0.09 \text{ m})^2 = r^2$$

$$r = \sqrt{(0.09 \text{ m})^2 + (0.09 \text{ m})^2}$$

$$r = 0.127 \text{ m}$$

Now, we can calculate each of the torques and add them together to find the net torque. Make sure to calculate the angle between $\vec{F}$ and $\vec{r}$ while their tails are coincident.

$$\begin{aligned}\tau_1 &= (28 \text{ N})r \sin(135^\circ) & \tau_2 &= (14.9 \text{ N})r \sin(135^\circ) & \tau_3 &= (16.1 \text{ N})r \sin(90^\circ) \\ \tau_1 &= 2.52 \text{ N} \cdot \text{m} \curvearrowright & \tau_2 &= 1.34 \text{ N} \cdot \text{m} \curvearrowright & \tau_3 &= 2.05 \text{ N} \cdot \text{m}\end{aligned}$$

Let counterclockwise rotation be the positive direction.

$$\begin{aligned}\Sigma\tau &= -\tau_1 + \tau_2 + \tau_3 \\ \Sigma\tau &= -(2.52 \text{ N} \cdot \text{m}) + (1.34 \text{ N} \cdot \text{m}) + (2.05 \text{ N} \cdot \text{m}) \\ \Sigma\tau &= \mathbf{0.870 \text{ N} \cdot \text{m}}\end{aligned}$$

10.2 Torque and Angular Acceleration

- Because force and linear acceleration are related to torque and angular acceleration, respectively, by a perpendicular distance from the origin, we can use Newton's second law to establish a relationship between torque and angular acceleration.

Definition 10.2.1 (Rotational Analog of Newton's Second Law)

Let all values reference some rigid body which rotates about a rotational axis z . Given moment of inertia I and angular acceleration α_z , let net torque be defined as follows:

$$\Sigma\tau_z = I\alpha_z$$

Example 10.2.2 (Rotating Machinery)

A machine part has the form of a solid uniform sphere with mass 220 g and diameter 4.50 cm. The part spins about a frictionless axle through its center, but at one point on its equator, it scrapes against metal, resulting in a friction force of 0.0200 N there.

- Find its angular acceleration. Let the direction of the sphere's rotation be the positive direction.
- How long will it take to decrease its rotational speed by 19.0 rad/s?

Continued on next page

Example (Rotating Machinery *continued*)

1) We can rearrange the formula for torque in terms of I and α_z . We can say that the torque is negative because it results from friction.

$$-\tau_z = I\alpha_z$$

$$\alpha_z = \frac{-\tau_z}{I}$$

$$\alpha_z = \frac{-fR}{\frac{2}{5}MR^2}$$

$$\alpha_z = \frac{-5(0.02 \text{ N})}{2(0.220 \text{ kg})(0.0225 \text{ m})}$$

$$\alpha_z = -10.1 \text{ rad/s}^2$$

2) Because we just solved for the angular acceleration, we can use an angular kinematic equation to find the time when $\Delta\omega_z = -19.0 \text{ rad/s}$.

$$\omega_z = \omega_{0z} + \alpha_z t$$

$$\Delta\omega_z = \alpha_z t$$

$$t = \frac{\Delta\omega_z}{\alpha_z}$$

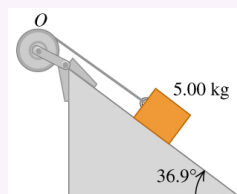
$$t = \frac{-19 \text{ rad/s}}{-10.1 \text{ rad/s}^2}$$

$$t = 1.88 \text{ s}$$

Example 10.2.3 (Flywheel on an Incline)

A block with mass $m = 5.00 \text{ kg}$ slides down a surface inclined 36.9° to the horizontal. The coefficient of kinetic friction is 0.27. A string attached to the block is wrapped around a flywheel on a fixed axis through O . The flywheel has a mass $m_f = 16.0 \text{ kg}$ and a moment of inertia

$I_f = 0.500 \text{ kg} \cdot \text{m}^2$ with respect to the axis of rotation. The string pulls without slipping a distance of 0.250 m from that axis.



- 1) What is the acceleration of the block down the plane?
- 2) What is the tension in the string?

Continued on next page

Example (Flywheel on an Incline *continued*)

1) Because we are only given the flywheel's moment of inertia and its mass, we need to solve for the radius using the corresponding moment of inertia equation (disk/cylinder).

$$\begin{aligned}\frac{1}{2}m_f R^2 &= I \\ R &= \sqrt{\frac{2I}{m_f}} \\ R &= \sqrt{\frac{2(0.5 \text{ kg} \cdot \text{m}^2)}{16 \text{ kg}}} = 0.25 \text{ m}\end{aligned}$$

The only force acting on the flywheel is a tension force from the rope. Thus, we can express the tension force in terms of known quantities. The unknown a will be factored out later.

$$\begin{aligned}\Sigma \tau_z &= I\alpha = I\left(\frac{a}{R}\right) = TR \\ T &= \frac{Ia}{R^2}\end{aligned}$$

Now, we can substitute this value for T into a Newton's second law equation to isolate a and solve.

$$\begin{aligned}\Sigma F_x &= mg \sin(36.9^\circ) - T - \mu_k mg \cos(36.9^\circ) \\ ma &= mg \sin(36.9^\circ) - \frac{Ia}{R^2} - \mu_k mg \cos(36.9^\circ) \\ ma + \frac{Ia}{R^2} &= mg \sin(36.9^\circ) - \mu_k mg \cos(36.9^\circ) \\ a &= \frac{mg(\sin(36.9^\circ) - \mu_k \cos(36.9^\circ))}{m + \frac{I}{R^2}} \\ a &= \frac{(5 \text{ kg})(9.81 \text{ m/s}^2)(\sin(36.9^\circ) - 0.27 \sin(36.9^\circ))}{5 \text{ kg} + (0.5 \text{ kg} \cdot \text{m}^2)(0.25 \text{ m})^{-2}} \\ \mathbf{a} &= \mathbf{1.45 \text{ m/s}^2}\end{aligned}$$

2) Now that we've solved for the acceleration a , we can solve for T using the equation we derived in part 1.

$$\begin{aligned}T &= \frac{Ia}{R^2} = \frac{(0.5 \text{ kg} \cdot \text{m}^2)(1.45 \text{ m/s}^2)}{0.25 \text{ m}} \\ \mathbf{T} &= \mathbf{11.6 \text{ N}}\end{aligned}$$

◇

10.3 Rotation About a Moving Axis

- If a rigid body's rotational axis is moving, we describe it using both its translational motion and its rotational motion.

$$K_{\text{tot}} = K_{\text{cm}} + K_{\text{rot-cm}}$$

$$K_{\text{tot}} = \frac{1}{2}M_{\text{tot}}v_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2$$

Rolling without Slipping

- When an object **rolls without slipping**, its point of contact with the surface it rolls on should be at rest. Thus, the velocity at the point of contact must be equal and opposite of v_{cm} .
- We generally say that static friction causes the rotation of the rolling object, but it does not affect the total energy of the system because it is at a minimum.
- Thus, for these cases, the following must hold true (given radius R , speed of the center of mass v_{cm} , and angular speed ω):

$$v_{\text{cm}} = R\omega$$

Important

Thus, the total kinetic energy of an object that rolls without slipping can be simplified. Let C be some constant associated with an object's moment of inertia.

$$K_{\text{tot}} = \frac{1}{2}M_{\text{tot}}v_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2$$

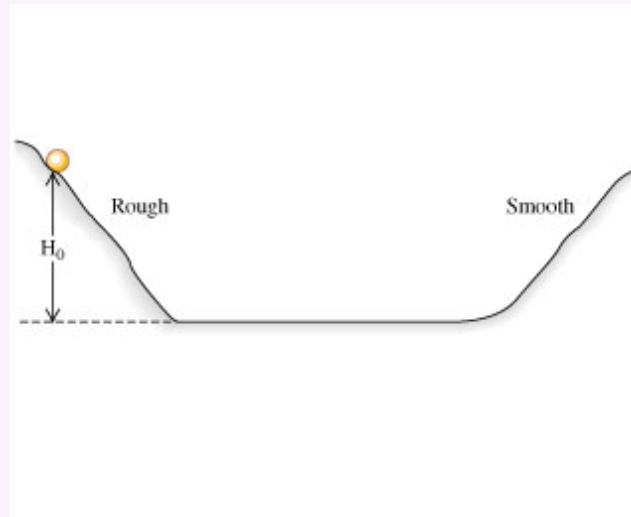
$$K_{\text{tot}} = \frac{1}{2}M_{\text{tot}}v_{\text{cm}}^2 + \frac{1}{2}(CM_{\text{tot}}R^2)\left(\frac{v_{\text{cm}}}{R}\right)^2$$

$$K_{\text{tot}} = \frac{1}{2}M_{\text{tot}}v_{\text{cm}}^2 + \frac{C}{2}M_{\text{tot}}v_{\text{cm}}^2$$

$$K_{\text{tot}} = \left(\frac{1+C}{2}\right)M_{\text{tot}}v_{\text{cm}}^2$$

Example 10.3.1 (Rolling Basketball)

A basketball, which can be approximated as a hollow sphere, rolls down a mountainside into a valley and then up the opposite side, starting from rest at a height H_0 above the bottom. The mountainside the basketball is released on is rough, preventing slipping. Meanwhile, the opposite side is smooth and frictionless. Neglect rolling friction and assume the system's total mechanical energy is conserved. How high, in terms of H_0 , will the ball go up the other side?



We can start by deriving an expression for H_0 using conservation of energy between the basketball's initial position and the valley.

$$\begin{aligned}
 mgH_0 &= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \\
 mgH_0 &= \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{3}mR^2\right)\left(\frac{v}{R}\right)^2 \\
 mgH_0 &= \frac{5}{6}mv^2 \\
 H_0 &= \frac{5v^2}{6g}
 \end{aligned}$$

Now, using the conservation of energy between the basketball's initial position and its final position and our derived H_0 , we can establish a connection between *only* H and H_0 . At the basketball's peak on the other side, it lacks translation kinetic energy but still has rotational kinetic energy from when it rolled down the rough part.

$$\begin{aligned}
 mgH_0 &= mgH + \frac{1}{2}I\omega^2 \\
 mgH_0 &= mgH + \frac{1}{2}\left(\frac{2}{3}mR^2\right)\left(\frac{v}{R}\right)^2
 \end{aligned}$$

Continued on next page

◇

Example 10.3.2 (Rolling Basketball *continued*)

$$\begin{aligned}
 mgH_0 &= mgH + \frac{1}{3}mv^2 \\
 g\left(\frac{5v^2}{6g}\right) &= gH + \frac{1}{3}v^2 \\
 \frac{5v^2}{6} &= gH + \frac{1}{3}v^2 \\
 H &= \frac{v^2}{2g} \\
 H &= \frac{3}{5}\left(\frac{5v^2}{6g}\right) \\
 H &= \frac{3}{5}H_0
 \end{aligned}$$

10.4 Work and Power in Rotational Motion

- Say that a tangential force acts on the rim of a pivoted disk. If the point on the rim moves by a distance of s :

$$\begin{aligned}
 dW &= F_{\text{tan}} ds \\
 dW &= F_{\text{tan}} R d\theta \\
 dW &= \tau_z d\theta
 \end{aligned}$$

- Now, by integrating both sides, we can reach a definition for work in rotational motion.

Definition 10.4.1 (Work in Rotational Motion)

Given torque τ_z , and angular positions θ_1 and θ_2 , let work be defined as follows:

$$W = \int_{\theta_1}^{\theta_2} \tau_z d\theta$$

- By taking the derivative with respect to time from both sides, we can also reach a definition for power in rotational motion.

Definition 10.4.2 (Power in Rotational Motion)

Given torque τ_z , and angular velocity ω_z about some rotational axis z , let power be defined as follows:

$$P = \tau_z \omega_z$$

10.5 Angular Momentum

- The rotational analog of a particle's momentum is its **angular momentum**, measured in $(\text{kg} \cdot \text{m}^2/\text{s})$.
 - Denoted L .
 - Its relationship to momentum is analogous to torque's relationship to force.

Definition 10.5.1 (Angular Momentum)

Given position vector $\vec{r}$ (relative to some reference point), and momentum $\vec{p}$, let angular momentum be defined as follows:

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$$

- Recall that the derivative of momentum with respect to time is force.
- Thus, we can also say that the *derivative of linear momentum with respect to time is torque from the net force*.

$$\frac{d\vec{L}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt}$$

$$\frac{d\vec{L}}{dt} = \vec{r} \times \vec{F}$$

$$\frac{d\vec{L}}{dt} = \vec{\tau}$$

Angular Momentum on a Rigid Body

- The total angular momentum of a rigid body is equal to the sum of all its particles' angular momenta.
- On a rigid body, we can express angular momentum in terms of angular speed.

$$L_{\text{tot}} = \sum_{i=0}^N L_i$$

$$L_{\text{tot}} = \sum_{i=0}^N r_i p_i = \sum_{i=0}^N r_i m_i v_i = \sum_{i=0}^N r_i m_i (r_i \omega) = \sum_{i=0}^N m_i r_i^2 \omega$$

$$L_{\text{tot}} = I\omega$$

Definition 10.5.2 (Angular Momentum on a Rigid Body)

On some rigid body, given moment of inertia I , and angular velocity $\vec{\omega}$, let angular momentum be defined as follows:

$$\vec{L} = I\vec{\omega}$$

- This definition supports the previous relationship we established between torque and the derivative of angular momentum with respect to time, as $d\vec{\omega}/dt = \vec{\alpha}$.

10.6 Conservation of Angular Momentum

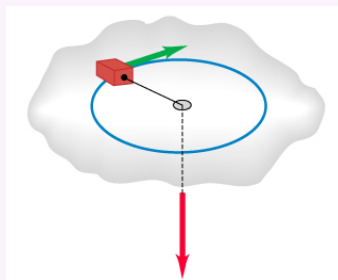
Definition 10.6.1 (Conservation of Angular Momentum)

We say that a system's total angular momentum is *conserved* if the system's net external torque is zero.

- Similarly to the conservation of momentum, the internal torques exerted between system particles change their individual angular momenta, but *do not impact the system's total angular momentum*.
- This is a direct consequence of Newton's third law.

Example 10.6.2 (Spinning Block)

A small block on a frictionless, horizontal surface has a mass of 2.50×10^{-2} kg. It is attached to a massless cord passing through a hole in the surface. The block is originally revolving at a distance of 0.300 m from the hole with an angular speed of 2.19 rad/s. The cord is then pulled from below, shortening the radius of the circle in which the block revolves to 0.150 m. Model the block as a particle.



- 1) Is the angular momentum of the block conserved?
- 2) What is the new angular speed?
- 3) Find the change in kinetic energy of the block.

1) **Angular momentum is conserved because the net force is just the rope's tension, which always points in the radial direction, parallel to the lever arm, meaning the net torque is always zero.**

2) Because we know that angular momentum is conserved, we can use the angular speed definition to solve for the new angular speed.

$$\begin{aligned}
 L_1 &= L_2 \\
 I_1 \omega_1 &= I_2 \omega_2 \\
 mr_1^2 \omega_1 &= mr_2^2 \omega_2 \\
 \omega_2 &= \omega_1 \frac{r_1^2}{r_2^2}
 \end{aligned}$$

Continued on next page

Example (Spinning Block *continued*)

$$\omega_2 = (2.19 \text{ rad/s}) \frac{(0.3 \text{ m})^2}{(0.15 \text{ m})^2}$$

$$\omega_2 = \mathbf{8.76 \text{ rad/s}}$$

3) We can use the previous angular speed we solved for to find the change in kinetic energy.

$$\Delta K = \frac{1}{2}m(v_2^2 - v_1^2)$$

$$\Delta K = \frac{1}{2}m(\omega_2^2 r_2^2 - \omega_1^2 r_1^2)$$

$$\Delta K = \frac{1}{2}((8.76 \text{ rad/s})^2(0.15 \text{ m})^2 - (2.19 \text{ rad/s})^2(0.3 \text{ m})^2)$$

$$\Delta K = \mathbf{1.62 \times 10^{-2} \text{ J}}$$

Example 10.6.3 (Parachutist on a Turntable)

A large wooden turntable in the shape of a flat uniform disk has a radius of 2.00 m and a total mass of $1.40 \times 10^2 \text{ kg}$. The turntable is initially rotating at 3.00 rad/s about a vertical axis through its center. Suddenly, an 80.0 kg parachutist makes a soft landing on the turntable at a point near the outer edge.

- 1) Find the angular speed of the turntable after the parachutist lands.
- 2) Compute the kinetic energy of the system before and after the parachutist lands.

1) About the rotational axis, there will be zero net torque after the parachutist lands, so angular momentum is conserved. Thus, we can use the angular speed definition to find the answer.

$$L_1 = L_2$$

$$I_1 \omega_1 = I_2 \omega_2$$

$$\omega_2 = \omega_1 \frac{I_1}{I_2}$$

$$\omega_2 = \omega_1 \frac{\frac{1}{2}m_t R^2}{\frac{1}{2}m_t R^2 + m_p R^2}$$

$$\omega_2 = (3 \text{ rad/s}) \frac{\frac{1}{2}(140 \text{ kg})(2 \text{ m})^2}{\frac{1}{2}(140 \text{ kg})(2 \text{ m})^2 + (80 \text{ kg})(2 \text{ m})^2}$$

$$\omega_2 = \mathbf{1.40 \text{ rad/s}}$$

Continued on next page

Example (Parachutist on a Turntable *continued*)

2) We can use the angular speed we just found and the definition for rotational kinetic energy.

$$K_1 = \frac{1}{2} I_1 \omega_1^2$$

$$K_1 = \frac{1}{2} \left(\frac{1}{2} m_t R^2 \right) \omega_1^2$$

$$K_1 = \frac{1}{4} ((140 \text{ kg})(2 \text{ m})^2) (3 \text{ rad/s})^2$$

$$\mathbf{K_1 = 1260 \text{ J}}$$

$$K_2 = \frac{1}{2} I_2 \omega_2^2$$

$$K_2 = \frac{1}{2} \left(\frac{1}{2} m_t R^2 + m_p R^2 \right) \omega_2^2$$

$$K_2 = \frac{1}{2} \left(\frac{1}{2} (140 \text{ kg})(2 \text{ m})^2 + (80 \text{ kg})(2 \text{ m})^2 \right)$$

$$\cdot (3 \text{ rad/s})^2$$

$$\mathbf{K_2 = 588 \text{ J}}$$



11 Equilibrium and Elasticity

11.1 Conditions for Equilibrium

- A particle is in **equilibrium** if it does not accelerate and has no tendency to rotate.

Definition 11.1.1 (Equilibrium Conditions)

We say that an object is in equilibrium if the following conditions on *any* point in the object are satisfied:

$$\Sigma \vec{F} = 0$$

$$\Sigma \vec{\tau} = 0$$

- When a rigid body at rest is evaluated to be in equilibrium under these conditions, we specify that it is in **static equilibrium**.

11.2 Center of Gravity

- Given that the decrease in weight force exerted across an object decreases at higher altitudes is negligible, we can say that the center of mass is the same as the **center of gravity (cg)**.
- We can calculate the torque due to weight force by assuming that a singular weight force acts on the object's center of gravity.
- We can prove this using our vector definition for center of mass:

$$\begin{aligned}\vec{\tau}_g &= \sum_{i=1}^N \vec{\tau}_i = \sum_{i=1}^N (\vec{r}_i \times m_i \vec{g}) = \sum_{i=1}^N (m_i \vec{r}_i) \times \vec{g} \\ &= \frac{\sum_{i=1}^N (m_i \vec{r}_i)}{\sum_{i=1}^N m_i} \times M_{\text{tot}} \vec{g} \\ &= \vec{R}_{\text{cm}} \times \vec{w}\end{aligned}$$

- The center of gravity of simple uniform objects is at their center. If additional forces applied, we can counterbalance them with forces using torque principles we learned in chapter 10.
- For more complicated objects, we can imagine an axis through a point of suspension. Now, the center of gravity must be on this axis for the net torque on the hanging object to be zero, so we can *suspend the object from another point, and the resulting intersection of suspension axes will yield the center of gravity*.
- Applying this to real life, an object with several points of support must have a center of gravity within the bounds of its supports. An object will be overturned if its center of gravity exits these bounds due to torque.
- Consequently, objects with larger footprints and lower centers of gravity are harder to overturn.

11.3 Rigid Body Equilibrium Problems

- We will cover problems with forces acting a singular plane, generally the xy -plane. Thus, only the following conditions will matter:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0$$

$$\Sigma \tau_z = 0$$

- Additionally, there are many possible reference points to calculate torque with respect to, so consider points that lead to simpler calculations.

Example 11.3.1 (Center of Gravity during a Plank)

A 1.8 m tall athlete measured that 66.0% of his weight was supported by his forearms while 34.0% of his weight was supported by his toes by maintaining a plank position on two scales. If the distance between his toes and the middle of his forearms is 1.53 m, how far are his toes from his center of gravity?

Like we established earlier, we can imagine a singular weight force w acting on the center of gravity. Additionally, using the weight percentages provided by the problem, we can express the normal forces in terms of the weight. Let L denote the distance between the athlete's toes and the middle of his forearms.

$$\Sigma \tau_z = -n_t R_t \sin(90^\circ) + n_f R_f \sin(90^\circ) + w \sin(0^\circ) = 0$$

$$-n_t R_t + n_f R_f = 0$$

$$-(0.34w)R_t + (0.66w)(L - R_t) = 0$$

$$0.66wL - wR_t = 0$$

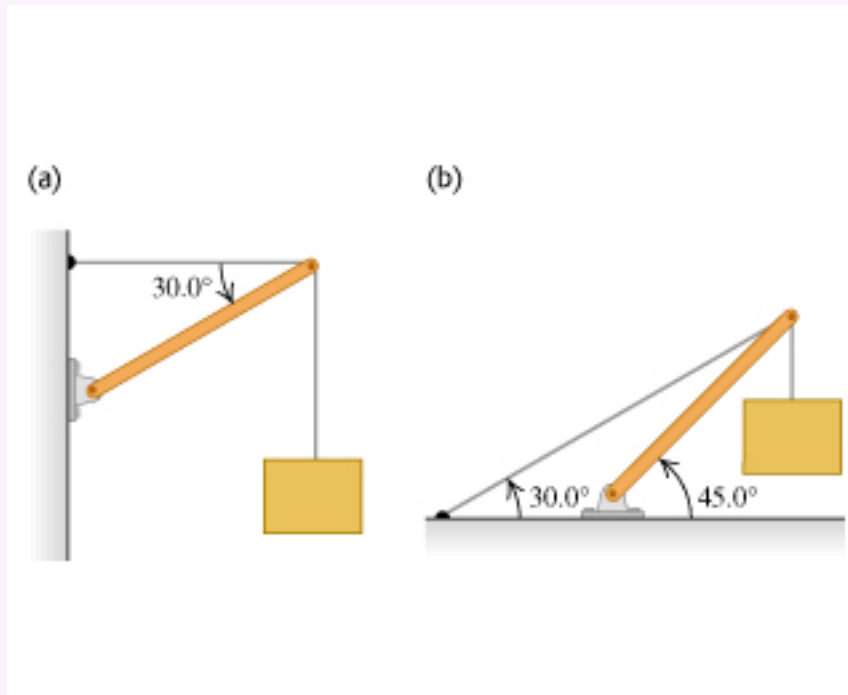
$$0.66wL = wR_t$$

$$R_t = 0.66L = 0.66(1.53 \text{ m})$$

$$R_t = 1.01 \text{ m}$$

Example 11.3.2 (Suspended Crate)

In each case, let w be the weight of the suspended crate. The strut is uniform and also has weight w .



- 1) In arrangement (a), find the horizontal cable's tension in terms of w .
- 2) In arrangement (a), find the magnitude of the pivot force in terms of w .

1) Let the point of contact with the wall be the reference point for the torque. We first define the vertical tension in terms of weight.

$$\Sigma F_y = T_C - w = 0$$

$$T_C = w$$

Now, the net torque equation is solely comprised of T_a and w .

$$\Sigma \tau_a = 0$$

$$\tau_T - \tau_g - \tau_C = 0$$

$$T_a L \sin(150^\circ) - w \frac{L}{2} \sin(60^\circ) - w L \sin(60^\circ) = 0$$

$$T_a = \frac{3w \sin(60^\circ)}{2 \sin(150^\circ)}$$

$$T_a = \frac{3\sqrt{3}}{2} w$$

Continued on next page

Example (Suspended Crate *continued*)

2) We can find F_{ax} in terms of w using our derived T_a .

$$\Sigma F_x = F_{ax} - T_a = 0$$

$$F_{ax} = \frac{3\sqrt{3}}{2}w$$

Similarly, we can find F_{ay} in terms of w using our derived T_C .

$$\Sigma F_y = F_{ay} - w - T_C = 0$$

$$F_{ay} = 2w$$

Now, we can find the magnitude of F_a .

$$F_a = \sqrt{F_{ax}^2 + F_{ay}^2}$$

$$F_a = \sqrt{\left(\frac{3\sqrt{3}}{2}w\right)^2 + (2w)^2}$$

$$F_a = \frac{\sqrt{43}}{2}w$$

3) In arrangement (b), find the horizontal cable's tension in terms of w .

4) In arrangement (b), find the magnitude of the pivot force in terms of w .

3) Like in the previous example, the tension of the vertical cord is equal to w . Now, if we let the point of contact with the ground be the reference point for the torque and calculate the angle between the strut and the diagonal cord (15°), we can find T_b .

$$\Sigma \tau_b = 0$$

$$\tau_T - \tau_g - \tau_C = 0$$

$$T_b L \sin(165^\circ) - w \frac{L}{2} \sin(45^\circ) - w L \sin(45^\circ) = 0$$

$$T_b = \frac{3w \sin(45^\circ)}{2 \sin(165^\circ)}$$

$$T_b = \frac{3w \sin(45^\circ)}{2 \sin(165^\circ)}$$

$$T_b = 4.09w$$

4) We can find F_{bx} using our derived T_b .

$$\Sigma F_x = F_{bx} - T_{bx} = 0$$

$$F_{bx} = 4.09w \cos(30^\circ) = 3.55w$$

Continued on next page



Example (Suspended Crate *continued*)

Similarly, we can find F_{by} in terms of w using our derived T_C and T_b .

$$\Sigma F_y = F_{by} - w - T_C - T_b = 0$$

$$F_{by} = 2w + 4.09w \sin(30^\circ)$$

$$F_{by} = 4.05w$$

Now, we can find the magnitude of F_b .

$$F_b = \sqrt{F_{bx}^2 + F_{by}^2}$$

$$F_b = \sqrt{(3.55w)^2 + (4.05w)^2}$$

$$F_b = 5.38w$$

Example 11.3.3 (Bicycle Wheel)

You are trying to raise a bicycle wheel of mass m and radius R up and over a curb of height h . To do this you apply a horizontal force $\vec{F}$ toward the cub.

- 1) What is the least magnitude of the force $\vec{F}$ that will succeed in raising the wheel onto the curb when the force is applied at the center of the wheel?
- 2) What is the least magnitude of the force $\vec{F}$ that will succeed in raising the wheel onto the curb when the force is applied at the top of the wheel?

For both questions, answer in terms of m , g , h , and R .

1) Because the bicycle wheel is being raised over the curb with a minimum force F , it rotates about the point of contact with the curb with a net torque of zero. When this happens, the only forces applying torque are $\vec{F}$ and gravity.

$$\Sigma \tau_z = mgR \sin(\theta) - FR \sin(90 - \theta) = 0$$

$$FR \cos(\theta) = mgR \sin(\theta)$$

$$F = mg \tan(\theta)$$

We can find the ratio of $\tan(\theta)$ in terms of R and h using the radial distance R to the corner of the curb as the hypotenuse of a right triangle. It follows that one of the legs is $R - h$, the vertical distance from the wheel's center to the curb's corner. Then, by the pythagorean theorem, the length of the other leg is $\sqrt{R^2 - (R - h)^2} = \sqrt{2Rh - h^2}$.

$$F = mg \frac{\sqrt{2Rh - h^2}}{R - h}$$

Continued on next page

Example (Bicycle Wheel *continued*)

2) The vertical distance from the top of the wheel to the curb's corner is $R_{\text{top-y}} = 2R - h$. Now, $\sin(\theta) = \frac{R_x}{R}$, while $\sin(\phi) = \frac{R_{\text{top-y}}}{R_{\text{top}}}$. Thus, we do not have to find the distance R_{top} because it will cancel out.

$$\Sigma \tau_z = mgR \sin(\theta) - FR_{\text{top}} \sin(\phi) = 0$$

$$FR_{\text{top}} \sin(\phi) = mgR \sin(\theta)$$

$$FR_{\text{top}} \left(\frac{2R - h}{R_{\text{top}}} \right) = mgR \left(\frac{\sqrt{2Rh - h^2}}{R} \right)$$

$$F(2R - h) = mg\sqrt{2Rh - h^2}$$

$$F = mg \frac{\sqrt{2Rh - h^2}}{2R - h}$$

11.4 Stress, Strain, and Elastic Moduli

- **Stress** is the measure of forces causing a deformation, generally in
- **Strain** refers to the deformations caused by stress.
- By **Hooke's Law**, the strength of a force is proportional to the strength of the deformations.

Definition 11.4.1 (Hooke's Law)

Elastic modulus represent the ratio of stress to strain.

$$\frac{\text{Stress}}{\text{Strain}} = \text{Elastic modulus}$$

- Now, Hooke's law is not a general law, as it only applies to smaller stresses and strains.
- Elastic modulus is dependent on the material of an object, not the shape or size.
- A material is **elastic** if deformations do not persist after the stress is removed.
- A material is **plastic** if deformations remain after the stress is removed.

Tensile and Compressive Stress and Strain

- An object is in **tension** when being stretched.
- **Tensile stress** refers to the forces deforming an object in tension, measured in pascals (Pa) or (N/m²).
- **Compressive stress** refers to the forms deforming an object through compression, also measured in (Pa).
- Pascals are also the unit of *pressure*.
- **Tensile strain** refers to the deformations an object in tension experiences.
- **Compressive strain** refers to the deformations a compressed object experiences.

Definition 11.4.2 (Stress)

Let $F_{\perp}$ be the force acting perpendicular to cross-sectional area A .

If $F_{\perp}$ acts outward:

$$\text{Tensile stress} = \frac{F_{\perp}}{A}$$

If $F_{\perp}$ acts inward:

$$\text{Compressive stress} = \frac{F_{\perp}}{A}$$

Note

Stress is a *scalar quantity* because $F_{\perp}$ refers to a magnitude.

Definition 11.4.3 (Tensile Strain)

Let Δl be the change in the deformed object's length and l_0 be the object's original length.

If Δl represents an elongation:

$$\text{Tensile strain} = \frac{\Delta l}{l_0}$$

If Δl represents a compression:

$$\text{Compressive strain} = \frac{\Delta l}{l_0}$$

Note

Strain is dimensionless because it is a ratio of two lengths.

Definition 11.4.4 (Young's Modulus)

Let Y be **Young's modulus** and let the other variables follow the definitions described in definitions (11.4.2) and (11.4.3):

$$Y = \frac{\text{Tensile stress}}{\text{Tensile strain}} = \frac{F_{\perp}/A}{\Delta l/l_0} = \frac{F_{\perp}}{A} \frac{\Delta l}{l_0}$$

Important

Definitions (11.4.1) and (11.4.4) also hold true for compressive strain and stress if the compressive stress is small.

- Some materials are resistant compressive stress but falter against similar tensile stress and vice versa.

11.5 Elasticity and Plasticity

- When observing larger stresses and strains, we graph stress as a function of strain.

-
- Strain is generally shown as the *percent elongation*.
 - The point on a stress versus strain graph where the slope is no longer constant is known as the **proportional limit** because that is the last point where Hooke's law applies.

14 Periodic Motion

- **Periodic motion** or **oscillation** refers to repetitive motions, such as the swinging of a pendulum.

14.1 Describing Oscillation

- In periodic motion, the equilibrium position is some arbitrary position where the displacement is chosen to be zero.
- **Restoring forces** move an object back to their equilibrium position after being displaced.
 - The spring force is an example of a restoring force.

Important

Oscillation can only occur if there is a restoring force.

- **Amplitude** is the maximum magnitude of displacement from equilibrium, measured in meters (m).
 - Denoted A .
- A **cycle** or complete vibration is entire round trip.
 - Starting at point A , a cycle would be $A \rightarrow -A \rightarrow A$.
- **Period** is the time needed to complete a cycle, measured in seconds (s), or sometimes seconds per cycle.
 - Denoted T .
- **Frequency** is the number of cycles per unit of time, measured in hertz (Hz) = s^{-1} , or sometimes cycles per second.
 - Denoted f .
- We covered period and frequency in our circular motion chapter, so you may also recall:

$$f = \frac{1}{T}$$

- **Angular frequency** is defined as follows:

$$\omega = 2\pi f$$

- The units are radians per cycle (rad/s).
- Like angular velocity, angular frequency is denoted ω .

Note

While angular frequency has the same definition as angular speed, their applications are different.

14.2 Simple Harmonic Motion

- In **simple harmonic motion (SHM)**, the net force is some restoring force F_x which is *directly proportional* to the displacement from equilibrium.

$$F_x \propto x$$

$$F_x = -kx$$

- A **harmonic oscillator** is an object in simple harmonic motion.

⚠ Caution

Not all periodic motions are simple harmonic, but many of them are *approximately* simple harmonic when the displacement is small as outlined in the previous chapter.

- Thus, the spring force is a good example of simple harmonic motion.
- In general, the restoring force constant is measured in N/m.
 - Denoted k .

Definition 14.2.1 (Acceleration in SHM)

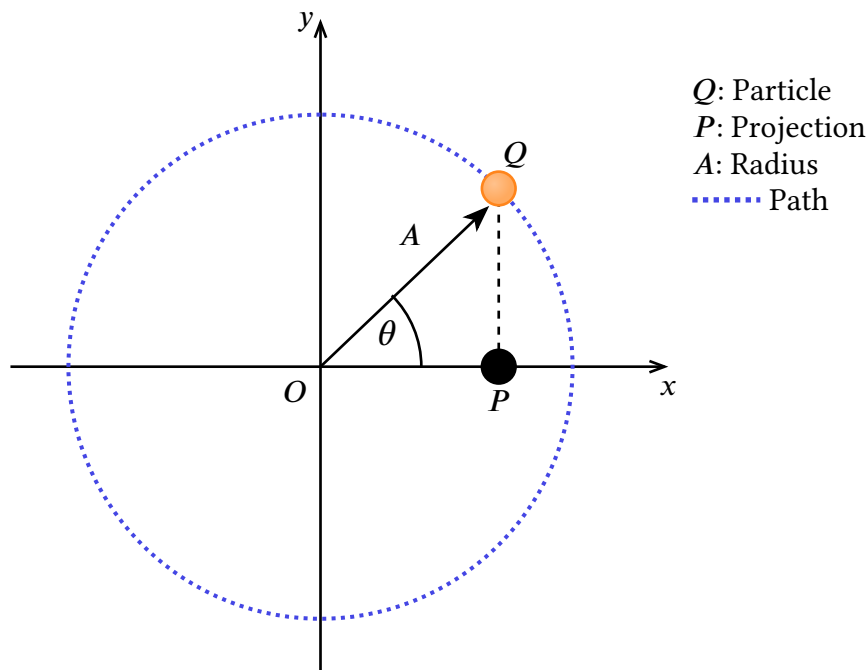
Given acceleration of an object in SHM a_x , restoring force constant k , and displacement x :

$$a_x = \frac{d^2x}{dt^2} = -\frac{k}{m}x$$



Relating Circular Motion to SHM

- Let Q be a particle in circular motion. Its projection P onto the x -axis will be in simple harmonic motion as Q completes revolutions.



- Using a **reference circle** and **phasor** (rotating vector), we can prove that Q can be used to model simple harmonic motion.

$$a_Q = \omega^2 A$$

$$R = A$$

- Now, the x -components of these values will refer to the acceleration and displacement, respectively, of the projection P .

Definition 14.2.2 (SHM Equations)

Let A be amplitude, k be the restoring force constant, ω be angular frequency, and m be the harmonic oscillator's mass.

$$x = A \cos(\theta)$$

$$a_x = -\omega^2 A \cos(\theta) = -\omega^2 x$$

$$\omega = \sqrt{\frac{k}{m}}$$



- This proves that the projection is in simple harmonic motion because P 's acceleration is directly proportional to $A \cos(\theta) = x$, the displacement.
- The definition for angular frequency comes from equating our new definition for SHM acceleration to our old one, $a_x = -\frac{k}{m}x$.

Remark

The greek letter, ω , is used for both angular speed and angular frequency because the quantities are equal. The time needed for Q to complete a revolution should be the same as the time needed for its projection P to complete a revolution.

Definition 14.2.3 (SHM Equations)

Let f and T be the frequency and period, respectively, in simple harmonic motion:

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$



Kinematics in SHM

Definition 14.2.4 (Kinematics in SHM)

Let ϕ be the phase angle, that is, the angle of the phasor at $t = 0$.

$$x = A \cos(\omega t + \phi)$$

$$v_x = -\omega A \sin(\omega t + \phi)$$

$$a_x = -\omega^2 A \cos(\omega t + \phi)$$



Graph	Description
	• SHM displacement graph with: $\phi = 0$ $\omega = 1 \text{ rad/s}$ • Oscillates between $-A$ and $+A$. • In other words, between $-x_{\text{max}}$ and $+x_{\text{max}}$.
	• SHM velocity graph with: $\phi = 0$ $\omega = 1 \text{ rad/s}$ • Oscillates between $-\omega A$ and $+\omega A$. • In other words, between $-v_{\text{max}}$ and $+v_{\text{max}}$.
	• SHM acceleration graph with: $\phi = 0$ $\omega = 1 \text{ rad/s}$ • Oscillates between $-\omega^2 A$ and $+\omega^2 A$. • In other words, between $-a_{\text{max}}$ and $+a_{\text{max}}$.

- This is further justification that our displacement and velocity functions are correct, as we verified this acceleration equation in the previous chapter.
- These equations also show how changes in A , k , and m impact a harmonic oscillator's motion.
 - A impacts the maximum values in SHM kinematics.
 - k and m impact the period and maximum values for acceleration and velocity.
- Now, if we are given an initial displacement and velocity, we can find A and ϕ by rearranging our equations.

$$\begin{aligned}\frac{v_{0x}}{x_0} &= \frac{-\omega A \sin(\phi)}{A \cos(\phi)} & x_0^2 + \frac{v_{0x}^2}{\omega^2} &= (A \cos(\phi))^2 + \frac{(-\omega A \sin(\phi))^2}{\omega^2} \\ -\frac{v_{0x}}{x_0 \omega} &= \tan(\phi) & x_0^2 + \frac{v_{0x}^2}{\omega^2} &= A^2 \\ \phi &= \arctan\left(-\frac{v_{0x}}{x_0 \omega}\right) & A &= \sqrt{x_0^2 + \frac{v_{0x}^2}{\omega^2}}\end{aligned}$$

14.3 Energy in SHM

- In simple harmonic motion, the total mechanical energy is conserved.
- As we've observed with springs in chapter 7, a harmonic oscillator reaches its maximum kinetic energy when at equilibrium.
- Meanwhile, a harmonic oscillator reaches its maximum potential energy when $|x| = |A|$, the amplitude.

Definition 14.3.1 (Total Energy in SHM)

$$E_{\text{SHM}} = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$$



14.4 Applications of SHM

- In the previous subchapters, we covered SHM in solely horizontal scenarios.
- However, we defined SHM to be any system with any restoring force acting directly proportional to displacement from equilibrium.
- By finding values of k in different scenarios, we can show that they model SHM.

Vertical SHM

- Say an object of mass m is hanging vertically from a spring with spring constant k . The object is at equilibrium when *the spring is stretched enough to balance the weight force*.
- Lets say that stretch distance is Δl , so:

$$k\Delta l = mg$$

- Now, let x be the distance from the equilibrium position. Using the previous equality we established, it follows that:

$$\text{Above equilibrium} \quad \Sigma F = k(\Delta l - x) - mg = -kx$$

$$\text{Below equilibrium} \quad \Sigma F = k(\Delta l + x) - mg = kx$$

- The object still oscillates with $\omega = \sqrt{\frac{k}{m}}$ despite being vertical.
- The main difference between horizontal and vertical SHM is the difference in equilibrium.

Angular SHM

- In angular SHM, we define a **restoring torque** similarly to a restoring force.
 - The constant κ is known as the **torsion constant**.

$$\Sigma \tau_z \propto \theta$$

$$\Sigma \tau_z = -\kappa \theta$$

$$\alpha_z = -\frac{\kappa}{I} \theta$$

- Additionally, we define angular frequency, period, and frequency similarly, albeit with the torsion constant and moment of inertia.

Definition 14.4.1 (Angular SHM Equations)

Let κ be the torsion constant:

$$\omega = \sqrt{\frac{\kappa}{I}}$$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{\kappa}{I}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{\kappa}}$$

- We also define angular displacement similarly to displacement.

Definition 14.4.2 (Angular Displacement in SHM)

Let ϑ be the angular amplitude and let the other variables be defined as in (14.2.2).

$$\theta = \vartheta \cos(\omega t + \phi)$$

- An example of this is a mechanical watch that keeps track of time using a balance wheel. A coil spring exerts a restoring torque proportional to the angular displacement of the wheel, returning it back to equilibrium.
- Here, the oscillations are used to keep track of time, essentially being an angular pendulum.

14.5 Simple Pendulum

- A **simple pendulum** is an idealized model where a point mass is suspended by a massless, unstretchable string.
- When released from either side of its straight-down equilibrium position, it oscillates along an arc of radius L , the length of the string.
- Now, let θ be the angle between the string and the equilibrium position of the string.
- Thus, the angle between w and w_y must also be θ because w is parallel to the equilibrium position and w_y is parallel to the string's current orientation.

$$\Sigma F_x = mg \sin(\theta) \quad \Sigma F_y = T - mg \cos(\theta) = 0$$

- Now, because $\sin(\theta) \approx \theta$ as $\theta \rightarrow 0$, the motion of a simple pendulum is *approximately harmonic*.

Definition 14.5.1 (Simple Pendulum Approximation)

Given a small angle θ and a displacement x from equilibrium, the restoring force for a simple pendulum can be approximated as follows:

$$F_\theta = -mg\theta = -\frac{mg}{L}x$$

- Here, the restoring force constant is approximately $\frac{mg}{L}$, which simplifies nicely in our previous SHM equations.

Definition 14.5.2 (Simple Pendulum Equations)

$$\omega = \sqrt{\frac{L}{g}}$$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{L}{g}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{g}{L}}$$

- However, if we want an exact value for T or have large angular displacements, we need to use a series.

$$T = 2\pi \sqrt{\frac{g}{L}} \left(1 + \frac{1^2}{2^2} \sin^2\left(\frac{\Theta}{2}\right) + \frac{1^2 \cdot 3^2}{2^2 \cdot 4^2} \sin^4\left(\frac{\Theta}{2}\right) + \dots \right)$$

- This shows why pendulums are great at keeping time, as changes in amplitude over time have little impact on the period.

14.6 Physical Pendulum

- A **physical pendulum** refers to a *real pendulum* using an extended body.
- Thus, the restoring force acts on the pendulum's center of gravity, causing a restoring torque at some angle θ .

$$\Sigma \tau_z = -mgd \sin(\theta)$$

- Again, we can use the approximation $\sin(\theta) \approx \theta$ when $\theta \rightarrow 0$ to simplify this equation.

$$\Sigma \tau_z = -mgd\theta$$

$$\alpha_z = -\frac{mgd}{I}\theta$$

Definition 14.6.1 (Physical Pendulum Equations)

$$\omega = \sqrt{\frac{mgd}{I}}$$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{mgd}{I}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{mgd}}$$

